

Project Report
Project 4

RECRUITAURA

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(December 2024)

Candidate's Declaration

We hereby certify that the work on the project entitled, "**Project Name - RecruitAura**", in partial fulfillment of requirements for the award of Degree of **Bachelor of Technology** in School of Engineering and Technology at BML Munjal University, having University Roll No. 210471, 210285, 210190 is an authentic record of our own work carried out during a period from February 2024 to May 2024 under the supervision of Dr. Anantha Rao and Dr. Atul Mishra.

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Supervisor's Declaration

This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

Faculty Supervisor Name: Dr. Atul Mishra

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Faculty Supervisor Name: Dr. Anantha Rao

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Abstract

RecruitAura depicts an effective turning point in technical recruitment, offering a breakthrough solution whereby an AI-powered and machine learning (ML) based platform is used to automate the process and boost efficiency. This can be attributed to the fact it was meant to solve the major challenges which hinder technical hiring such as extensive time requirement, shortage of resources and lack of experienced engineers that can conduct interview. RecruitAura enables automation on the critical areas of interview by leveraging the cutting-edge NLP as well as DL technologies that include real-time coding analysis, speech-to-text conversions, sentiment analysis, and non-verbal cue analysis to mention a few. On this platform, therefore, the entire logistic side of recruitment is simplified, in addition to enriching the quality of evaluations, thereby improving the comprehensive and fair assessment process. The coming test runs have been a source of hopefulness, as there has been a clear sign of improvement in efficiency, candidate experience, and interviewer workload. With future view RecruitAura is going to spread its technological range and plans to extend its geographic reach, with the goal of remodeling the global recruitment standards and creating a more unified and user-friendly hiring space.

Acknowledgement

We are highly grateful to Dr. Anantha Rao, Assistant Professor and Dr. Atul Mishra, Assistant Professor, BML Munjal University, who were our advisors & supervisors. They not only provided their valuable advice but also helped us to execute this project successfully over five months time frame, i.e. Feb to May 2024.

The competence and skill set of Dr. Atul Mishra in areas of deep learning, and natural language processing not only manifested itself as significant resources but gave rise to the inspiration that propelled our project development all the way. Implementation of his specific directions was the key to successful understanding of technical details that were smarting at us at some point.

Equally, we also owe much to Dr. Anantha Rao for his aid in the concepts related to software engineering. His proven ability to systematically convey these principles and guide us towards the project realization were of the great help to our understanding and practical implementation of the project.

We would like to thank Dr. Mishra and Dr. Rao for the practical suggestions and the never-ending motivation that they continue to give us. Their prompt advice and the discussions to keep up our faith and spirit became the key elements that we had for our project going on well and our team spirit.

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1. Introduction

The software industry keeps revolutionizing at a high speed which has resulted in many difficulties in finding technical talents. Long software interviews detract the delivery, in an attempt to find the skilled ones, most of which are occupied by hiring experts. The inefficiency is also aggravated by the small size of the number of those who can conduct these interviews which is due to the fact that these people are exposed to their role in the operations of the organization and are overwhelmed with responsibilities. Taking these issues into account, we are calling for the creation of RecruitAura—an incredible platform, which is based on the use of AI and ML technologies to change the technical interview process to the extent that it can be fully automated.

The main goal of RecruitAura is to enable quick and smooth, fully remote technical interviews. As a result, we are able to drastically reduce the time and resources used for the evaluation process, among other things. The platform has an inbuilt NLP (Natural Language Processing) and DL (Deep Learning) algorithms and therefore can offer swift coding assessments, speech-to-text conversion of interview responses and insight extraction from answers. This traditional method relied on questionnaires and essays, and today, psychometric tests have evolved, eliminating errors through sentiment analysis, cheat detection, and the analysis of nonverbal cues, which in turn helps recruiters create more comprehensive and quality candidate assessment.

Driven by the urgent need to rise to the severity of technical hiring within the software sector, especially during a time characterized by the high demand for technical skills, is where RecruitAura focuses its efforts. Organizations need to make a balance between quickly finding and choosing appropriate applicants and providing a safe and merit-based evaluation course. RecruitAura boosts a company's capability of executing technical interviews better and makes the hiring decisions more judicious with all the testings and observations done by the engineers from the wide pool of the organization.

RecruitAura will change how tech recruitment can be done for the better by taking away the mundane work while boosting the quality of the hiring process. With this invention the burden of engineers will be deleted, the evaluation of the candidates will be instant and objective and a remote, accessible and universal interview process for worldwide candidates will be created. Such a mechanism maintains a high level of integrity and equity in the measuring and assessment process which is to the advantage of the employers by simplifying job hiring process and is in line with the overall goals of growing the tech industry by efficiently getting the best suited skills in place at the right time.

As opposed to traditional platforms which focus only on challenges in coding, RecruitAura does an overall assessment of candidates by the real-time speech-to-text conversion and analysis of interviews, information extraction, and sentiment analysis to understand the technical skills as well as soft skills.

Outside of this, deep learning-based cheat detection and optional non-verbal feature analysis will allow for a more detailed assessment of the suitability and the fidelity of the participants. The platform also uses standardized rubrics for in-depth candidate assessment, advanced features for detecting cheating, and offers 24/7 availability, which facilitates the hiring process for business engineers around the world who follow different time zones. The smart engine's holistic approach to technical recruiting stands out among the other existing solutions, as it facilitates greater depth and efficiency level.

2. Introduction to Project

RecruitAura is envisioned as a transformative tool in the recruitment industry, designed to streamline the hiring process by interfacing organizations, interviewers, and candidates through a centralized platform. This system not only facilitates the outsourcing of interviews by companies but also offers professional interviewers a channel to utilize their skills for monetary benefits. Simultaneously, it simplifies the interview process for candidates, ensuring a smooth progression through each step.

2.1 Overview

In addition to saving time and streamlining the process of surfacing the best candidates as possible, RecruitAura will also create an opportunity for evaluating the capability of recruiting individuals. The main strength of RecruitAura is the possibility for organizations to create filters which then work through submitted resumes retrieve only the most suitable candidates. This targeted approach sets the set for the fair and complete evaluation of applicants, thus extending the quality of the hiring process. This individualized technique also demonstrates that there is no miss in evaluating the candidates, making the decision making process robust.

Apart from that, the platform serves as the source of profit for interviewers who can use their industry knowledge to conduct the interviews and get a passive income. Herewith, RecruitAura is a platform for experienced professionals who are looking for ways to put their knowledge and interview skills into play outside their regular work.

Candidates would get a unique benefit from this groundbreaking technology. RecruitAura provides a simple and straightforward UI that lets applicants easily navigate their applications and interview periods. This systematic way eliminates the stress and insecurity that normally occurs during the job search process, since it provides a more structured way of moving through the recruitment process.

2.2 Existing System

Karat

The Karat platform has the reputation of an advanced technical interviewing system which enables companies to substantially improve the level, throughput and diversity of their technical hiring pipeline. The platform offers an array of intelligent products and services, including 24/7 interviews, bubble-free environment, and reliable interviews with highly knowledgeable professionals. Karat's method emphasises human specialists joining with technology to intelligently place into the hiring process and present proven results wherever the recruitment goes . Company places great emphasis on

developing assessment tools, which are predictive, fair, and engaging, thus distinguishing it from other solutions and ensuring a consistent and systematic experience for candidates and employers alike. The fact that Karat is driven by the desire to develop an improved hiring process is revealed by its construction of job competency models for structured and validated jobs while linking questions for job interviews to expectations and competencies of workplace roles. The Karat candidates get the chance to interact with live professional interviewers who will adapt their algorithm to candidates' needs and preferences. This way, the Karat Interview Engineers offer a fair and predictive interview experience and all the candidates walk away happy, and their organization will hire the right talent efficiently. Generally, Karat is deserving of being recognized for the innovation it has brought to the job market by making the technical interviews more accurate and fair, and enjoyable making a high rating in the tech hiring industry.

Intervue.io

Intervue.io, a company with the achievement of digital transformation, has been exceptional in the modernization of technical hiring. The site underscores its 1-stop solution that unites recruiters with technology talent on-the-fly. By dint of the fact that Intervue.io is tasked with changing the traditional face-to-face-interview and aims to the one and only way the interviews are conducted globally. The company delivers a technical assessment program, which act as a test tool through technical questions and provide with team rooms to promote collaboration during the interview. Furthermore, Intervue.io has highlighted the use of shared IDEs and whiteboards to enable brainstorming in different languages. This puts forward the concept of improving the feedback cycle between developers and HR managers, and eventually respond to this feedback by making superior hiring decisions that are based on data. Furthermore, the platform provides organizations the chance for them to work with top engineers from the ages to solve the evaluation of candidate and this consequently enables the speeding up of talent acquisition process. Overall, Intervue.io has received an appreciation for its unique idea in tech hiring and its effort in improving the recruiting experience for all the parties involved.

Kariyer.net

Whereas a long time ago Kariyer.net was merely a place where employers listed jobs, today it is a multipurpose workplace that provides services such as employer branding, virtual recruitment events, and training for tech talent. Kariyer.net, the flagship online employment service in Turkey, is known for the unique job listings offered and adding some more features to come up with a key output. The company applies AI-driven job matching which is essential in perfecting job seekers with the relevant areas of job placement. This is made possible by the power of artificial intelligence in generating a logical and accurate solution. Through the use of this site, Kariyer.net is able to identify trends in job descriptions, candidates profiles, and historical hiring data. This, in return, helps to deliver the best job

recommendations possible. Additionally, the website carries out the activities such as employer branding and online job fairs. They make it possible for businesses to display their conduct and vacant positions and promote communication between employers and those job-seekers who live far from work location. Responding to the rise for technical ability, the platform through their technical job board for tech roles.

2.3 User Characteristics

RecruitAura is engineered to serve three primary user groups: The professionals of HR specialties and technical recruiters, personnel with relevant engineering expertise, and applicants. Each group interacts with the platform with specific needs and expectations.

Technical Recruiters and HR Professionals

Profile: These users typically manage the logistical aspects of the hiring process within organizations.

Needs:

- **Automation Tools:** They necessitate the use of automation to carry the core procedures uninterrupted with time like scheduling candidates, tracking them and preliminary assessments.
- **Integration:** The system based on integration needs to interfere well with other HR processing systems to provide an effective flow of work.
- **Reporting Features:** Sophisticated reporting tools on your interview schedule feed becomes essential for any successful decision making process.

Behavior: These are power users who engage with the platform regularly. It is, henceforth, a requirement of a platform to work with agility and ease to be able to handle multiple tasks at a time.

HR Professionals

Profile: Human Resources managers and recruiters who oversee the end-to-end hiring process, ensuring that the company attracts and retains top talent efficiently and effectively.

Needs:

- **Streamlined Workflow Management:** A platform that centralizes all hiring activities, including application review, interview scheduling, and feedback collection, ensuring a smooth and structured process.
- **Comprehensive Candidate Insights:** Access to detailed candidate profiles, including transcripts, skills analysis, and behavioral assessments, to make informed hiring decisions quickly.
- **Integrated Scheduling Tools:** Simplified tools for scheduling and coordinating interviews, addressing conflicting schedules, and sending automated reminders to candidates and hiring managers.
- **Data-Driven Insights:** Analytics dashboards to assess recruitment trends, measure key performance indicators (KPIs), and identify areas for improvement in the hiring process.
- **Compliance and Fairness Tools:** Features to ensure interviews are unbiased and compliant with legal hiring practices, with consistent evaluation metrics and equal opportunity checks.

Behavior: HR professionals like Marcus use the platform daily during recruitment cycles, particularly when filling urgent or high-volume positions. They rely on the system for efficient multitasking, managing communication between stakeholders, and ensuring the process is fair and transparent. They value tools that save time and promote professionalism.



Marcus Lee

Age – 35
Hometown – Singapore
Education – Singapore University
Occupation – Human Resources Manager

“Efficiency is doing things right; effectiveness is doing the right things.”

GOALS

- Find qualified candidates efficiently to fill open positions within the company.
- Ensure a smooth and professional interview process for both candidates and hiring managers.
- Improve the company’s reputation as an employer of choice by providing a positive interview experience.

CHALLENGES

- Sorting through numerous job applications to identify qualified candidates is time-consuming.
- Coordinating interviews between candidates and hiring managers can be challenging, especially when schedules conflict.
- Ensuring consistency and fairness in the interview process while providing valuable feedback to candidates.

DESCRIPTION

Marcus is an experienced HR manager working in a large corporation. He is responsible for recruiting top talent for the company’s various departments. Marcus is always looking for efficient ways to streamline the hiring process and find the best candidates for open positions.

Figure 1 Example persona for an HR professional

Experienced Engineers

Profile: Experts available for technical assessment of candidates who showcase their talents in the facilitation of the recruitment process.

Needs:

- **User-Friendly Interface:** A simple but user-friendly interface that guide them to focus entirely on the job of analyzing their candidates without the fear that they will either get off track or be overwhelmed by the technical complexities of the platform.
- **Efficient Communication Tools:** Tools that will be using to communicate with candidates and ensure that communication is clear and the interviewees are briefed.
- **Customizable Evaluation Metrics:** Among activities and feedback, it can personalize it for different technical roles by modifying evaluation metrics and feedback forms to meet the required ones.

Behavior: It is the norm for them to use the platform on an intermittent basis while mostly focusing on the days they are on schedule to do their interviews. They work fast and prepare all the data on task.



Arun Kumar

Age - 42
Hometown - Delhi
Education - NSUT Delhi
Occupation - Freelance IT Consultant

“Success is not just about the opportunities you find, but the ones you create with your expertise, determination, and relentless pursuit of excellence.”

GOALS

- Find contract opportunities with reputable companies that require his expertise.
- Showcase his skills and experience effectively.
- Maximize his earning potential by securing high-paying contracts.

CHALLENGES

- Finding new clients and contract opportunities in a competitive market.
- Convincing potential clients of his capabilities.
- Negotiating favorable contract terms and rates with clients.

DESCRIPTION

Arun is a seasoned IT consultant with over 15 years of experience in the industry. As a freelancer, he's always on the lookout for new opportunities to expand his client base and take on challenging projects.

Figure 2 Example persona for an Experienced Interviewer

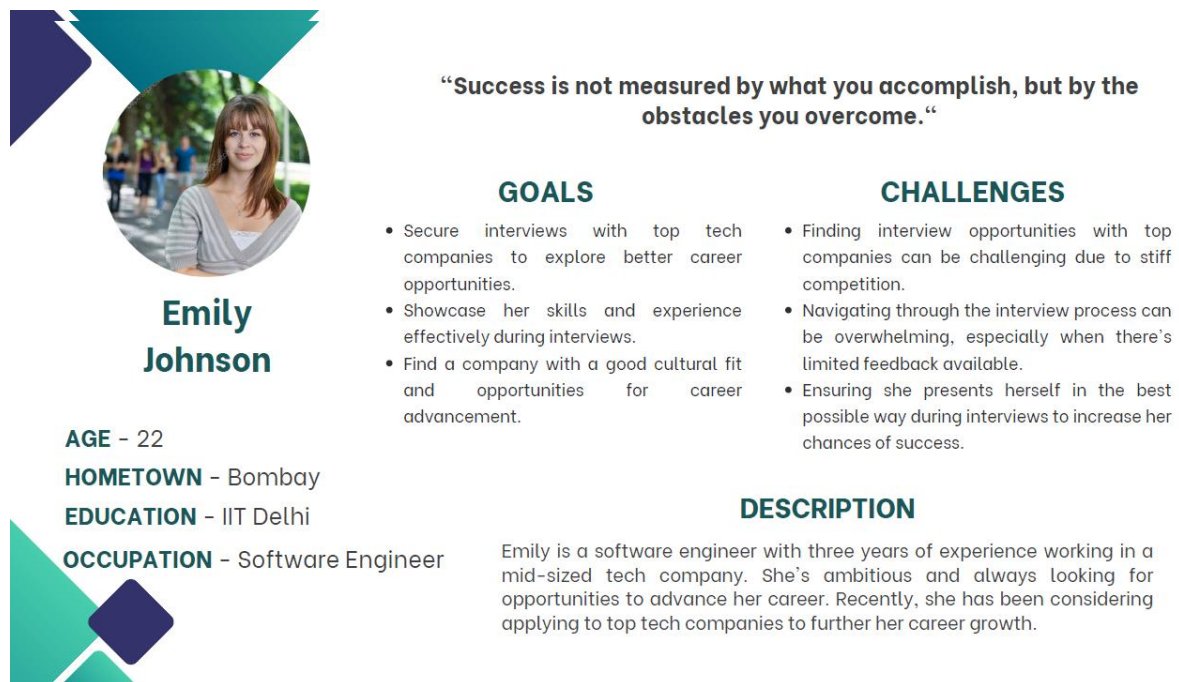
Job Candidates

Profile: The job seekers benefit from this platform to apply for jobs, attend job interviews and get feedback.

Needs:

- **Accessibility:** The platform has to be available on different devices that candidates can access wherever they are and use it.
- **Responsive Support:** Fast customer support with an intention to assist in overcoming any challenges during the registration, application, or interview stage.

Behavior: The engagement of the platform by them is very unpredictable but it can be pretty high while they are at the job searching stage. Simplicity and dependability are two major factors that must be considered if one wishes to present a healthy environment for the users.



The figure is a candidate persona card for Emily Johnson. It features a circular profile picture of a woman with brown hair. To the right of the photo is a quote: "Success is not measured by what you accomplish, but by the obstacles you overcome." Below the photo, the name "Emily Johnson" is written in bold. To the left of the name, there are four lines of text: "AGE - 22", "HOMETOWN - Bombay", "EDUCATION - IIT Delhi", and "OCCUPATION - Software Engineer". To the right of the name, there are two columns of text. The first column is titled "GOALS" and lists three bullet points: "Secure interviews with top tech companies to explore better career opportunities.", "Showcase her skills and experience effectively during interviews.", and "Find a company with a good cultural fit and opportunities for career advancement." The second column is titled "CHALLENGES" and lists three bullet points: "Finding interview opportunities with top companies can be challenging due to stiff competition.", "Navigating through the interview process can be overwhelming, especially when there's limited feedback available.", and "Ensuring she presents herself in the best possible way during interviews to increase her chances of success." Below these columns is a section titled "DESCRIPTION" which contains a paragraph: "Emily is a software engineer with three years of experience working in a mid-sized tech company. She's ambitious and always looking for opportunities to advance her career. Recently, she has been considering applying to top tech companies to further her career growth." The entire card is decorated with teal and purple geometric shapes.

Emily Johnson

AGE - 22
HOMETOWN - Bombay
EDUCATION - IIT Delhi
OCCUPATION - Software Engineer

GOALS

- Secure interviews with top tech companies to explore better career opportunities.
- Showcase her skills and experience effectively during interviews.
- Find a company with a good cultural fit and opportunities for career advancement.

CHALLENGES

- Finding interview opportunities with top companies can be challenging due to stiff competition.
- Navigating through the interview process can be overwhelming, especially when there's limited feedback available.
- Ensuring she presents herself in the best possible way during interviews to increase her chances of success.

DESCRIPTION

Emily is a software engineer with three years of experience working in a mid-sized tech company. She's ambitious and always looking for opportunities to advance her career. Recently, she has been considering applying to top tech companies to further her career growth.

Figure 3 Example persona for a candidate

2.4 Feasibility Study

The feasibility study assesses the technical, economic, and operational sides of the business and establishes whether RecruitAura actually addresses a need in our job market scenario or it is just an idea that needs more work.

Technical Feasibility

Robust Platform Requirements: The scope includes the development of a strong, easily scalable web platform that integrates cutting-edge functions users within all the stakeholders groups and supports large traffic and data accuracy.

Integration Capabilities: Installing tools that facilitate the swift integration of new platform components with existing HR systems and databases to maintain uninterrupted operations.

Security Measures: Designing and using of high-level security methods to save users confidential data and preserve their privacy.

Economic Feasibility

Cost-Benefit Analysis: A financial plan showing the startup and running expenses for the facility spent as compared to forecasted profits, which would help to appraise the economic proportions of the project.

Market Analysis: Measuring the demand of the indicated platform among the vast industries; and the competition world.

Revenue Models: Many possible sources of revenue are being sought e.g. services fees to organizations, subscription fees for premium features and other commission-based models which pay to interviewers.

Operational Feasibility

User Adoption Strategy: Efficient marketing strategies which segment the target audience to improve the highest level of adoption amongst them.

Customer Support and Maintenance: Construct of a coverage and well-organized support system which will provide assistance to users, address their concerns fast, increase satisfaction level among the users and upgrade the reliability of the platform.

Long-Term Sustainability

Scalability Plans: Plans for scaling the platform taking into account the user numbers growth and massive data volume.

Continuous Improvement: Frequent process of feedback collection and review from users and streamlining processes with technology.

Finally, by developing the long-awaited solution in the recruitment industry, RecruitAura will fill the voids which exist now in the hereto inexistent regimes of efficiency, transparency and user responsiveness. This platform is created to tackle the emerging demands of the job market and help organizations accomplish their missions successfully together with candidates and interviewers. Proceeding cautiously with quality based planning, list of tasks, and execution strategies, RecruitAura is here to transform the area of recruitment practices to worthy standards.

3. Literature Review

With technology representation across many fields, the area of online education via virtual recruitment has experienced an influx of the use of complex technologies fabricated to improve the operations of this field. Consequently, it becomes essential to look into digital technologies that are currently implemented when sitting online tests and taking part in virtual recruitment procedures and outline how such modernizations afford both security measures and interpersonal evaluations through facial and vocal analyses. The exploration encompasses a range of studies highlighting the application of 360-degree security cameras, artificial intelligence in pre-interview screening, as well as the fact that facial and vocal recognition technologies have to be used carefully to reach a comprehensive understanding of what the candidate is capable of and the emotional state of the candidate.

The usage of automated video interviews (AVIs) for weeding out job candidates is on the rise. On the other hand, there is a glaring absence of research that focus on exploring the reliability, validity and the generalization of the assessments made during this process. Moving forward with the deficiency of valuation of divergent personality characteristics, Hickman and the team have determined AVIs which get to analyze verbal, paraverbal, and nonverbal behaviors from video interviews to check the Big Five personality traits. The results indicate that AVIs trained on interviewer reports displayed inconsistent results, they showed convergent and discriminant validity, used conceptually relevant predictors, and handled academic achievement prediction, which in turn was well done. Another debatable issue for doing self-report avis is that these are bias on the part of the interviewee and thus may have weak reliability and validity. Displaying an example of an unguided conversation which was very helpful [1].

In addition, the use of AI technology for screening processes is one of the topics that the research studies such as "Job Pre-Interview System with Artificial Intelligence" and " AI-based Behavioural Analyser for Interviews/Viva." The first one explains an AI system which does pre-interviewing by using semantic analysis and natural language processing to determine the applicant suitability. The latter research explores the potential of AI in decoding candidate's non-verbals, which are considered the best indicators for attributes such as personality trait and state of mind during the interview. In this respect, the ability of AI powered by deep learning to imitate human behaviour indicates that this AI would be much more effective in the assessment of interpersonal skills in the virtual interviews; thus, it would give the possibility of a deeper evaluation process [2][3].

The human tendency to misread facial expressions and body language is amplified in the virtual environment. With online education and the process of virtual recruitment on its rise, the requirement of effective candidate monitoring and evaluation approaches has become indispensable. The research presented in " Students Online Exam Proctoring: A Case Study Using 360-Degree Security Cameras" emphasises on having the ability of 360-degree security cameras to be used in online examination

settings. Based on this technology, the monitoring scope will be significantly increased which may result in higher security and chance to disable opportunities of cheating. Consequently, it is created to facilitate in reducing the amounts of stress and performance anxiety as they are the usual areas where students don't do well whilst using the surveillance methods. This examination scenario creates a relaxed and less invasive exam atmosphere for students participating in it [4].

Visual analytics does not only build graphical interfaces, but it also facilitates human machine collaboration by using both voice and visual inputs to interpret emotions and ideas. Shuai Peng et al. improved the trainability, stability and performance in face recognition by activating the combination of activation functions and trainable parameters on the Inception-ResNet architecture. Moreover, their work showed superior results on several datasets [5]. Jia Xiang and Gengming Zhu also used Multi task cascaded convolutional networks (MTCNN) for face detection and micro expression recognition to make more accurate analysis of nonverbal communication. It renders human computer interaction more subtle by capturing subtle cues from facial expressions [6].

Building this, Na Li and colleagues extended this to study eye and attention tracking and uncover perceptual and attentional behaviours in varying structural environments to further understand human visual behaviour[7]. Together, these studies highlight the broad potential of visual analytics to enhance human computer and inter human interactions. Visual analytics, represented by Peng's deep learning advancements which show how AI can become more responsive to human emotions, as well as Li's eye movement and attention tracking research, continue to promise safety, personalization, and deeper engagement in technological processes.

Furthermore, candidate's vocal analysis becomes another key element in the interview process. Through the changing of the accent, tone, volume and intonation, the vocal cues become unnoticeable and the responses given by the candidate towards the questions and the comfort he or she has when talking about certain topics are easily revealed. This improves the interviewer's capability of judging the fit of the candidate to his position. While discussing about speech emotion recognition (SER) technology, Akçay and Öguz highlight the progress of classifiers like convolutional networks and recurrent neural networks and dire difficulties like dataset production and feature extraction [8]. While, Wani deals with the figure of Speech Emotion Recognition (SER) in human-computer interactions which is developed through the SER system designs. Also, improved emotional models and robust classifiers are suggested in order to accommodate diverse datasets, and reflect the quality of interview interfaces. Additionally giving us an indepth review of existing SER systems [9].

Voice analysis is also critical in maintaining the integrity of interviews to determine if multiple speakers are involved in audio recordings. Accurate detection of overlapping speech is possible even with pre-trained pyannote.audio diarization model which effectively segments and labels audio into distinct speakers [10]. GPU acceleration supported by PyTorch makes this possible for real time audio analysis

[11]. These technologies help to make the interview process more trustworthy, by eliminating external help for candidates, and preventing personal responses from being influenced.

As in parallel the bias detection systems are crucial for promoting inclusiveness and fairness by analysing dialogue to detect emotional inconsistencies, or language containing bigotry. Other tools like spaCy do the Named Entity Recognition (NER), syntactic parsing, and pattern detection to determine whether bias is present [12]. Advantageous to impartial evaluation are sentiment analysis, powered by nuanced NLP pipelines, that reveal tone and emotional anomalies. These systems enable merit based decision making and guide the interviewer to think about their language use.

Summarisation systems attempt to address the problem of long candidate documents by condensing resumes, reports and research papers to short, actionable summaries. We see the facebook/bart-large-cnn model generates high quality summaries, with contextual accuracy while keeping document length low [14]. PDFMiner extracts raw text from PDFs for preprocessing, and Hugging Face Transformers brings model integration much easier [15][16]. For instance, FPDF further enhances usability by producing well formatted, shareable summaries which let you uniformly and simply evaluate documents [17].

If you use the Applicant Tracking System (ATS) it automates the screening process by looking at resumes against job descriptions to find matches and gaps. Sentence BERT embeddings allow us to perform precise numerical comparison of resumes to job descriptions using cosine similarity to measure how aligned they are [18][19], while TF-IDF identifies the most important terms from a job description. PyPDF2 extracts textual data for processing, and spaCy identifies key skills and qualifications, to support accurate and efficient resume evaluations [20][12]. The ATS allows for candidates to receive actionable feedback to ensure they are writing their resumes to be aligned with specific roles and saves recruiters massive time.

Finally, attire detection systems approve or disapprove of a candidates look during virtual interviews if they do not adhere to dress codes. A real time object detection model, YOLO, can identify clothing categories at incredible speed and with incredible accuracy [21]. PyTorch optimises model deployment for efficient processing [15][11] and Hugging Face Transformers make it easy to fine-tune for specific requirements. These systems offer standardisation of evaluation of professionals' presentation while being inclusive, removing subjectivity and bias in the process.

The integration of advanced technologies in virtual recruitment and online assessments marks a pivotal transformation in how candidates are evaluated and monitored. AI-driven solutions such as Automated Video Interviews (AVIs), facial and vocal recognition, and natural language processing are redefining traditional recruitment by enhancing efficiency, security, and objectivity. These technologies allow deeper insights into candidates' interpersonal and emotional traits while addressing concerns like bias

and reliability, ultimately enabling more informed decision-making in educational and professional environments.

These advancements also signal a broader shift in online assessments, aiming to improve fairness and elevate traditional standards of candidate evaluation. As technologies supporting interview techniques continue to evolve, their incorporation into everyday practices will set new benchmarks for candidate management and assessment procedures across diverse settings. The future of virtual recruitment lies in balancing automation with human oversight, creating systems that are reliable, inclusive, and transformative for both individuals and organizations.

3.1 Comparison

This section gives a complex analysis of the key features, advantages, and disadvantages of similar products, thus helping us to understand our competitive stand and gather a further knowledge on possibilities for creating of new and innovative product or improvement.

Table 1 Critical analysis of existing products

Aspect	RecruitAura	Karat	Kariyer.net	Intervue.io
Innovation in Assessment	✓	✓		✓
Technology Utilization	✓			✓
FAANG Hiring Managers	✓	✓		✓
Support for Non-Traditional Backgrounds	✓	✓		
Efficiency in Hiring	✓	✓	✓	✓
Cheat Detection	✓			
Availability in India	✓	✓	✓	✓

3.2 Objectives of Project

For RecruitAura to successfully combat the difficulties inherent in hiring technical talent within this industry during this period of rapid software industry evolution, specific objectives have been set ranging from the improvement to absolute transformation of the technical interviewing process. These objectives are aimed at addressing candidate and business needs, guaranteeing that recruitment flow is simple, fair and smooth. Here are the key objectives for our project:

- **Enhance Interview Efficiency and Accessibility:** The primary mission of RecruitAura is the implementation of AI and ML technologies which allows the technical examination procedures to be more effective and more available for candidates. The platform creates opportunity to conduct interviews via radio, television or the internet. There you can do a smoothly conducted diagnostic medical interview, and thus, easily avoid the need to do an in-person interview and decrease the costly aspects of conventional interviewing.
- **Innovative Candidate Evaluation:** RecruitAura will explore the use of the NLP and DL methods to introduce features including real-time coding reviews, speech-to-text conversion for interview responses. together with the sentiment assessment. This leads to a more refined understanding while generating meaningful insights, which is a distinguishing factor from traditional evaluation methods as real-time polls are commonly used during debates.
- **Reduce Resource Burden on Companies:** Through automation of a good chunk of the interview process, RecruitAura will help human resources people to stiffe their workload, especially those who are very experienced such as the engineers who are often tasked with the interviews. Instead of small tasks, these highly capable employees will devote their capabilities to their primary responsibilities that will benefit the entire production of the company.
- **Fair and Unbiased Evaluations:** RecruitAura aims to have impartial and bias-free hiring of job applicants as one of critical goals. This rather extends the development of an AI system where the cheat detection system and analysis-based systems for retention of the interview process can work effectively. Besides, the platform's advance analytics will be helpful in finding more objective decision making in the hiring process.
- **Global and Inclusive Access:** This interface is emphasized on around the clock availability, and therefore companies and engineers operating across varying time zones can use interview services whenever they have is convenient. Due to the world wide adoption, this expands the resorts to adopt the inclusive hiring practice, allowing them to engage with talent world wide.
- **Continuous Improvement and Innovation:** The RecruitAura will constantly be advanced in AI and ML applications in a way to improve its capabilities. This type of change encompasses the implementation of advanced technologies, which have a more reliable performance, and the implementation of new systems following evolving industry standards and practices.

These goals fall into line with the principal idea of the process of technical recruitment transformation which is making it more simplicity orally, democratization and globalization and finally decent tech opportunity matchmaking in tech industry.

4. Methodology

4.1 Introduction to Languages

RecruitAura is a multi-dimensional application developed from a combination of powerful program languages which are well-known for their capabilities in the various domain where the application is being operate.. The main programming languages are Javascript, Python, and SQL, which constitute a well-structured backbone of the web application and the back-end part of the product.

React.js

It was picked up for its effectiveness and the fact that it operates around the virtual DOM element. js enables ScoringPro to operate with large data sets without any performance loss. It rewards designed a better commuting by creating a very responsive and interactive interfaces. It can be very much time-saving and simplifies the development process for UI by letting draw on common functions and arranging the state management more efficiently. Decreases the number of expensive components of DOM by using bundles to achieve the desired updates, which enhances the performance and the experience of the user.Minimizes the number of costly DOM manipulations by batching updates, improving performance and user experience. Provides JavaScript programmers an opportunity to develop HTML structures right within the code, which boosts the productive development of the code and improves the coding readability.

Python

Python has been favored by RecruitAura for the backend system for the implementation of its simplicity and the set of rich libraries. It is commonly preferred for its agile nature during startup startup in the cases where time-to-market is a very important aspect. Being free from obfuscated syntax and code readability in its design philosophy makes Python the programming language that simplifies maintaining, scaling up and developing complex stuffs, such as RecruitAura. Python utilizes the transparent and provisioned codes that facilitate instantaneous execution and easy maintenance. This outperforms other backend languages which require longer coding time and have hectic maintenance process. Offers plenty of help with backend development by using frameworks such as Flask and Django and libraries that are dedicated to manipulating data and recognizing the machines. Smooth interoperability with other technologies in terms that connect with diverse APIs and data sources.

SQL

SQL (Structured Query Language) brings about a vital function in RecruitAura handling the relational system at API tier, which is MySQL system. It provides sophisticated data management tools for running complex queries and this in turn is one of the major advantages because it is used for the entire span of platform, from user management through interview scheduling and results processing. SQL offers tools that allow for querying, sorting and the alteration data, which is the main area of an interviewing oriented platform. Provides support functionalities to carry even complex commands, which maintain the data integrity and consistency among operations. SQL databases are among the most scalable ones and also stand on high security platform, hence attracted to those applications that want to operate 24/7 and store sensitive types of data.

4.2 Supporting Languages/Packages

Flask

Implementing this micro-framework is perfect for those kind of projects where custom and complicated backend logic is needed, but not heavy-weight tools. This is true for the RecruitAura's aimed for this micro-framework. Flask has wide library base and integrates with other Python library modules seamlessly. This fact makes it a good choice in development of such applications as simple ones as well as complex ones. It can be assembled very easily and works with few complications, thus learning and launching your website is just a matter of clicks. There is no dependency conflicts, which makes the tools selection process simpler for developers by default. It advantages from huge community back up support and overwhelming stock of helpful tools for troubleshooting and further extension.

Python Libraries

NumPy and Pandas

A strong aspect is the ability to manage the numerical data effectively. Besides being a critical element, it provides high level data structures together with operation tools that make it easier to carry out complex data manipulation tasks. Designed for effectiveness and eligibility to manage big data loading in a short time and with great efficiency. It provides a wide range of features, including reshaping, merging, combining, cutting up and generalizing the data to get desired results.

SciPy

While NumPy acts as the complement to all this, new additions like modules for optimization, regression, signal processing, etc. can be considered for complex computational analytics. Gives a tool-

set of techniques to deal with numerical issues in integration, differentiation, optimization, and many others.

scikit-learn

One of the principal features of Machine Learning approaches utilized within RecruitAura, which provide for insights into predicting possible scenarios and data mining. Covers wide range of well-known ml algorithms like clustering, classification and regression among others. - Also offers specific ml models which work better for clustering, classification or regression features. It also very friendly with other Python libraries so one can be applied together, which can offer huge help on data science field.

spaCy

Wide range of features for the complex natural language processing, among others text-to-understanding of data required for RecruitAura. Such evaluation can be done faster and more reliably than by involving humans, especially in processing large amounts of text. Provide the word tokenization, nern, pos tagging and dependency parsing facilities for deeper analysis of interview focus.

nltk

Utilized for a variety of text processing tasks within RecruitAura. It enables the text splitting into sentences and words and is principal for the exclusion of usual language parts which, in fact, are quite useless in text analysis.

gensim

Based on unsupervised semantic modeling of plain text, it is essential for topic modeling within RecruitAura. It is efficient in discovering abstract topics from collection of large text. Assists in the management and the conversion of text data into a form that can be easily processed for the subsequent analysis.

textstat

The library is specialized for the evaluation of the readability and complexity of text is RecruitAura. Thus, it helps in the understanding and processing of data for RecruitAura. It allows for the fast and precise evaluation of the text readability and grade levels, thus, the speed and the reliability of the text evaluation are increased without any human involvement. It gives detailed information like syllable counts, sentence counts, and other readability indices, which are important for academic and content-specific analyses.

TensorFlow

A remarkable library for deep learning that allows machine learning practitioners to develop and train complicated models quickly and efficiently, which is vital for the tasks such as sentiment analysis within RecruitAura. Provides tools and libraries to develop and train advanced neural networks with ease, which are applicable to a wide range of tasks such as image recognition to sequence prediction. TensorFlow which can run on both CPUs and GPUs, is thus a tool that enables scalable and efficient model operations, which are necessary for the processing of large datasets.

dlib

Dlib is appreciated for its practical and uncomplicated way of machine learning and is applied to develop machine learning and face recognition models which are also useful for image processing. Provides tools for creating machine learning models that are both fast and have a low memory footprint, which is perfect for real-time applications. Provides a diverse set of algorithms for the feature extraction from images and other data types, thus, boosting the performance of the machine learning models.

pyannote.audio

The speaker diarization and audio processing tasks are the specialties of this library, where it identifies different speakers in audio recordings. Robust pre-trained models are provided to process conversations to extract 'who spoke where' with high accuracy. Overlapping speech is handled efficiently by the library and combined with GPU acceleration, audio data quickly and precisely segmented in real-time applications. It is therefore indispensable for monitoring equitable distribution of dialogue in interviews.

PyTorch

An implementable machine learning framework that is versatile enough to support different machine learning models. It supports the deployment of pre trained models and helps achieve computational efficiency. Torch is an important tool for building, training and deploying neural networks trivially for big data and for tasks like speaker diarization and real-time analysis. Because it's designed in a modular fashion, it can be easily integrated into almost any AI pipeline and its performance and scalability are increased.

PIL (Python Imaging Library)

Preprocessing image data is a very important task and PIL is the best tool to be used for resizing and normalising, as well as other types of format compatibility with deep learning models. It cleans and makes input data consistent so it is ready for analysis by the YOLO model. PIL is a foundational part of the system because this preprocessing step is vital to getting reliable object detection and classification while performing attire analysis.

PyTorch Lightning

Being an advanced module in PyTorch, PyTorch Lightning helps us in the development and management of deep learning models technically with tools of efficient training, logging, and checkpointing. The structured workflow makes working with devices easy and it also allows you to work seamlessly with a YOLO-Small model to train and deploy. The platform achieves consistent results with minimal overhead by automatically training in Lightning's automated training loops and through its modular design. It is a critical component of the attire detection system.

PyPDF2

A wonderful library for reading and extracting text from PDF documents is PyPDF2. It enables accurate scraping of content from resumes and job descriptions made available as PDF. Producing raw data by PyPDF2 that is ready for further analysis by handling text preprocessing efficiently.

FPDF

The content is converted by FPDF into a clean and professional PDF document. It guarantees proper formatted processed text ready to be shared by panels or archived for future use. Because of the flexibility and approachability that FPDF gives us, we are able to make our content as functional as can be and as aesthetically pleasing as possible.

Through ensuring that these languages and the technologies are all integrated, RecruitAura will be the seamless, efficient and scalable platform that dynamically meets these high speed environments today.

Integration and Other Technologies

WebRTC

WebRTC is a basic technology that gives high-quality real-time audio and video interactions through web browsers, eliminating the need for any additional installations of software. This quality makes WebRTC very ideal to conduct live interviews and interactive discussions on platforms like RecruitAura. Secure, end-to-end encrypted protocols ensure that all communications are confidential and integrity is maintained between interviewers and candidates, thereby fostering trust. Using the elements of intimacy and interaction as found in face-to-face interviews, WebRTC allows one to transcend geographical boundaries and enjoy co-presence despite all physical world separations.

Besides video communication, RecruitAura uses Firebase to enhance collaborative text-oriented capabilities, especially in chat interfaces and shared coding environments. With Firebase, collaborative text is immediately updated in real time for all participants. For video conferencing, Firebase validates

the unique room identifiers used in P2P interactions and hence creates a fluid and secure environment for dialogues.

The architecture of the platform depends on Firebase Firestore and Firebase Storage. This makes it possible to manage data efficiently. Firestore ensures that all collaborative contributions such as chat messages, code modifications, etc. are instantly synchronized and stored while Storage allows file sharing with size constraints that will give optimal performance. Apart from enhancing functionality, such integrations ensure that the platform is user-friendly and reliable.

Through the use of WebRTC for live interactions, as well as Firebase for data management regarding real-time performance, the Recruit Aura is available as a full-scope online interview and collaborative activities framework for candidates and interviewers to provide meaningful communication away from complex technology, thus secure yet efficient and highly accessible.

Rapid API

The Code Compiler API from Rapid API is included to enable that code snippets developed for coding challenges and interviews can be compiled and run properly through the candidates. This API is compatible with a wide range of programming languages hence there is no need to set different coding environment for different candidates. It stresses fast assessment of code, a fast response to participants, which is mandatory for tasks that imply skills in programming as a primary parameter. Code tests and challenges in the RecruitAura interface are run through the API and this is a plus since it means most of the connection to the user interface is smoothly done.

Whisper API

Relies on Whisper API to transcribe interviews, making the AI-based monitoring process more comprehensive. Through this process, the monitoring system could document and analyze interviews. This tool comes in handy for accessibility assistance and as a preparation mechanism for NLP-driven data insights. Provides high-quality voice recognition feature supporting various languages and different local dialects aiming at making RecruitAura reach a wider global audience. Beyond facilitating the recording of interview and conversation, it is the crucial component that allows for other analysis such as sentiment analysis and keyword extraction to be carried out. Whisper API platform is capable of communicating with the audio files recorded from the interviews in the course of real time. The written text that comes along with a transcript can be used by the HR managers and the interviewers for review and further analysis.

The specific technologies and APIs that are thoughtfully selected and integrated into the RecruitAura system are designed to support the objective of WFH interviewing which is providing a comprehensive,

efficient, and user-friendly interviewing platform. Use of each technology makes the platform not only to be enhanced regarding its specific functionalities, but also the whole operation will run in a smooth, secured and efficient way as well as in various industries that need professional talents in technical recruitment.

4.3 Constrains

The management of RecruitAura's development and operations is a complex process, which involves a number of different constraints that must be managed well so as to maximise the efficiency of the platform in different areas. Some of the main obstacles can be put into technical, resources, regulations and usability categories and each of them increase the difficulty in technical design, implementation and platform scaling.

Technical Constraints

System Performance: The platform should be able to deal with a lot of non-stop incoming data records, especially then there is more people to interview at the same time. It needs additional back-end systems for the data processing and data storage deficient any latency or downtime.

Scalability: The increasing numbers of user base and the amount of conducted interviews at the same time will influence the rate of system scaling to be efficient regarding the increased loads. This relates to the addition of server speed, database performance, and network capacity to achieve normal user experience.

Security: RecruitAura processes huge volumes to personal and confidential data are therefore security becomes a critical issue. The security of the platform remains on the focus by utilizing the best practices in security, including the encryption of data, using secure authentication methods, and implementing measures that mitigate common security risks such as SQL injections, cross-site scripting (XSS), and data breaches.

Compatibility: The platform must be functional beyond a single platform and across different types of devices, operating systems, and browsers platforms to reach all user groups. This is achieved by means of the method of multiple tests along with adopting the technique of optimization which is designed by keeping all the screen sizes and technology in consideration.

Resource Constraints

Budget: The significant difference in funds that are available to us result in the decisions which we have to make about where the minimal resources we have should be applied to different functions like

software development, marketing, server costs, and maintenance. Values of proportionality are the firestone of balanced allocations of expenditures, the ones that allow for the growth and innovation as well as for the operational efficiency.

Expertise: The level of the skills possessed by experts is a vital point since these include programmers, machine learning, and cybersecurity experts, among other professionals. Hiring and holding talent and key specialists may be challenging and can affect the technology uptake and innovation rates within the platform.

Time: In many cases the time needed passes under the yoke of several sources such as customers' appetites, the requirements of investors and a multitude of high-tech businesses. We should employ reasonable-demand management and agile development methods to ensure there are no delays while still maintaining at a very high-quality level.

Regulatory Constraints

Data Privacy: Adherence to international data protection regulation, for instance, the GDPR in Europe and CCPA in California, is a factor affecting the way user data is captured, maintained, and manipulated. Meeting those requirements entails the continuous tracking of regulations in place and the timely and appropriate adopting of system processes ensuring compliance.

Accessibility Standards: The platform must be user friendly and accessible to enable the economically disadvantaged people use it. This implies to be WCAG adherent (Web Content Accessibility Guidelines) and implementing features such as screen reader support, keyboard navigation, and ample contrast.

Employment Laws: Due to the fact RecruitAura job application is used in hiring, it must meet all the exclusion laws and regulations which secure the candidate rights and freedoms during the recruitment process.

Usability Constraints

User Experience: Obviously such program necessitates a simple and user-friendly interface, the fact that users may have different technological experiences posing a problem as well. The interface design has to be quite simple but at the same time functional making it without the user getting engaged easily.

Adaptability: This platform should respond to various user requirements as well as individual feedback - this calls for a highly-compliant design, where any changes, updates, or integrations can be incorporated into the existing system without causing serious disruptions.

Support: Offering functional speedy assistive solutions is crucial not only to fix user problems but to also cultivate user satisfaction. Which is through equipping support staff with a lot of resources to back them up e.g. deployment of help desk which is together with tutorial videos that aid user experience.

To remove all the constraints, a well-planned and balanced strategy is required which includes innovative design and pro-active management. Through properly handling the various issues, RecruitAura will be able to construct a proper platform that is scalable, secure, and which is user-oriented and addresses the ever changing and an aggressive hiring landscape.

4.4 Use Case

To effectively communicate how RecruitAura enhances the interview process for different stakeholders, we have made story boards that show specific scenarios within the platform. These storyboards show the systems and advantages received by the users by means of tracing their whole way of life.

Storyboard 1: Company's Hiring Procedures

Initial Need: A Company would need to employ new staff members yet it has no time to go through the interviewing process. It is the first panel that capture company's situation where it encounters its first challenge.

Solution Introduction: An employee proposes adopting RecruitAura, an external service by he staff members to find relevant and qualified candidates. Also, RecruitAura is considered as a solution for their needs.

Performance Review: The company obtains an in-depth report on the applicant by the interviewer and a job-related feedback from RecruitAura. This empowers the platform, therefore, making it a good tool to avoid being scammed.

Hiring Decision: Further, the company considers a well-structured interview conducted with RecruitAura's help where the interviewee is asked questions and the interviewer evaluates the candidate and then company make a decision. This last panel aims at highlighting the successful realization (by the platform) the objective (of the platform).



1
A company wants to hire new employees, but they do not have time to perform interviews.



2
One of the employees suggest Interview Pro, where the company can hire qualified professionals to take the interviews in place of the company.



3
The company receives a report on the performance of the applicant by the interviewee



4
The company makes an informed decision and hires the applicant based on the feedback given by the interviewee, all thanks to Interview Pro.

Figure 4 Storyboard: Company hiring

Storyboard 2: Experienced Professional

Seeking Opportunities: Let's look at Max, a career-oriented individual, who is pounding into the extra income idea. He wants to capitalize on his skills and keep his focus on the matter at hand. This frame highlights the introduction of the main character, Max and sheds light on his goals.

Discovering RecruitAura: On his friend's advice, Max gets introduced to RecruitAura, which is a platform where he can become a rather an interviewee. This highlights the platform as a place to give experts like Max a chance.

Active Participation: Max creates a free account at RecruitAura and then proceeds with the interviews for major companies as a demonstration. This panel brings to light the fact that talking with real people in the industry and getting presents on the platform is stress free.

Reward and Recognition: After numerous interviews were being performed, Max can deliver his feedback regarding candidates with the company and indicatively obtain his remuneration which all can be possible with RecruitAura. This last canvas showcases the thrilling experience for interviewers, culminating the full journey through the platform.



Meet Max, a working professional with 15 years of experience, he's looking for ways to earn some extra income without learning any new skill.



Max's friend suggest him to become an interviewer at Interview Pro.



Max registers at Interview Pro as an interviewer, and starts taking interviews on behalf of top companies.



Max provides his feedback about applicants to the company, and receives payment for a job well done, all thanks to Interview Pro.

Figure 5 Storyboard: Experienced Professional

Storyboard 3: Candidate's Journey

Initial Frustration: Congratulations on meeting Sarah who has been in the job market for so long and now is getting at the end of the job interview with great anticipation. This panel addresses these challenges as they impact job seekers. Understanding the root causes is crucial in addressing homelessness effectively.

RecruitAura Introduction: Sarah becomes familiar with the RecruitAura platform during a job application when a company invites her to the interview through this tool. This way, RecruitAura becomes a single platform where all the requirements for efficient hiring are available.

Seamless Experience: Impressed by One-round thorough assessment on RecruitAura Sarah sees that answers of a job interviewer are realistic and the process itself seems really simple. This illustrates how the website makes it simple for residents and especially visitors to have a hassle-free experience.

Successful Outcome: Sarah is offered and accepts the position after the company's successful access to RecruitAura, raising the question whether there is a cause connection between the tool and the company's ability to make quick hiring decisions.



1
Meet Sarah. She's tired of giving multiple rounds just to reach to the interview round for a job at a company.



2
A company she applied at, informed her about her interview getting scheduled at Interview Pro.



3
Sarah is immediately impressed by the experience, where she just have to go through one round, where the interviewer assessed her on all aspects.



4
Sarah soon gets a job confirmation from the company based on her interview, without any hassle, all thanks to Interview Pro.

Figure 6 Storyboard: Candidate's Journey

The story boards are aimed at unravelling RecruitAura from different points of the view. They elucidate how the company will permit employers doing the hiring to operate in an efficient way, how Max, a professional, will appreciate the flexible income opportunity owing to the app, and how Sarah, a candidate, will delight in the seamless screening process. Each storyboard links the user's start point with the end point in RecruitAura exploration, highlighting RecruitAura's integral role in changing again the traditional interview practice.

4.5 Data Flow Diagrams

The Data Flow Diagrams (DFDs) for the RecruitAura visualize the information flow pattern and present the systematic processes that user types go through on the platform. These diagrams depict the relationships between the interviewing staff, the companies, and the activity behind the scenes, showing the path on which data is exchanged and controlled within RecruitAura.

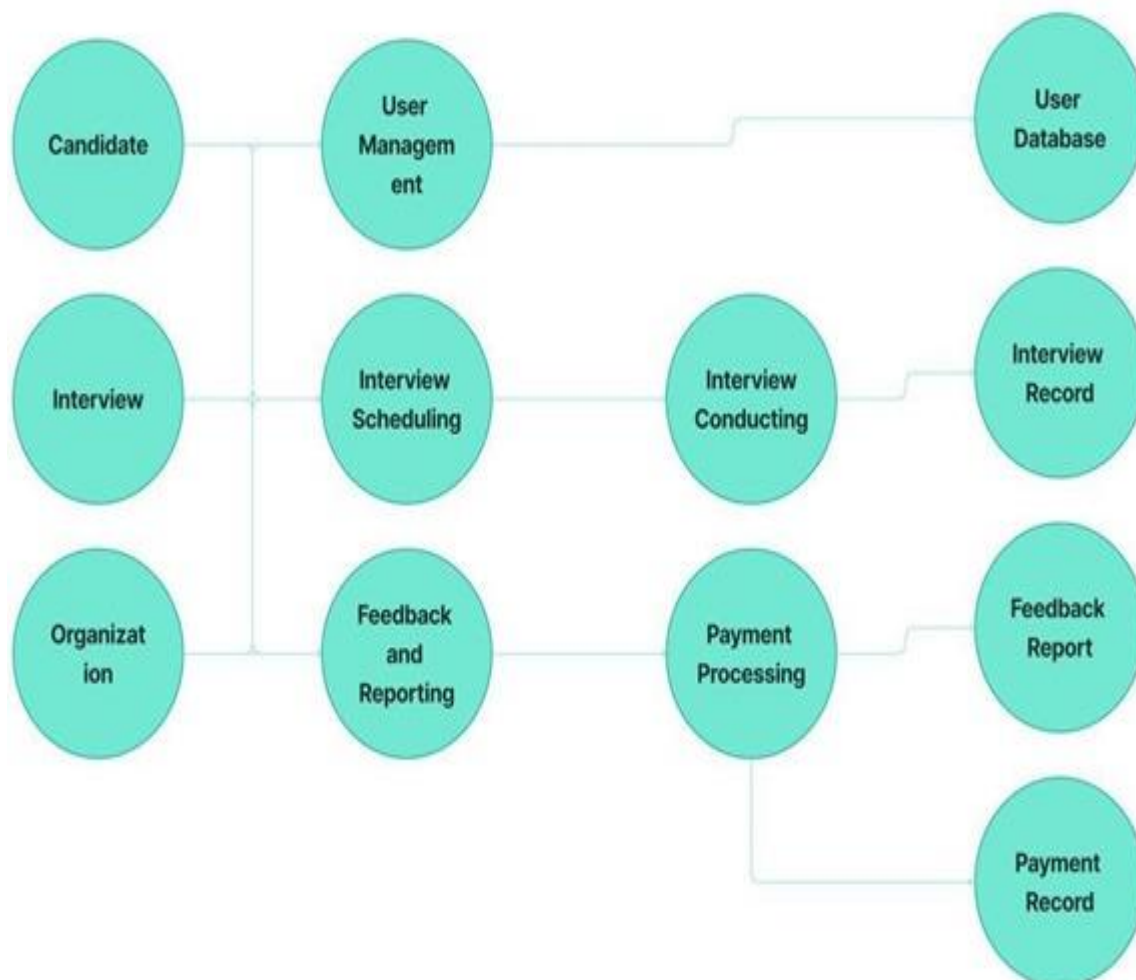


Figure 7 Data Flow Diagram Level 0

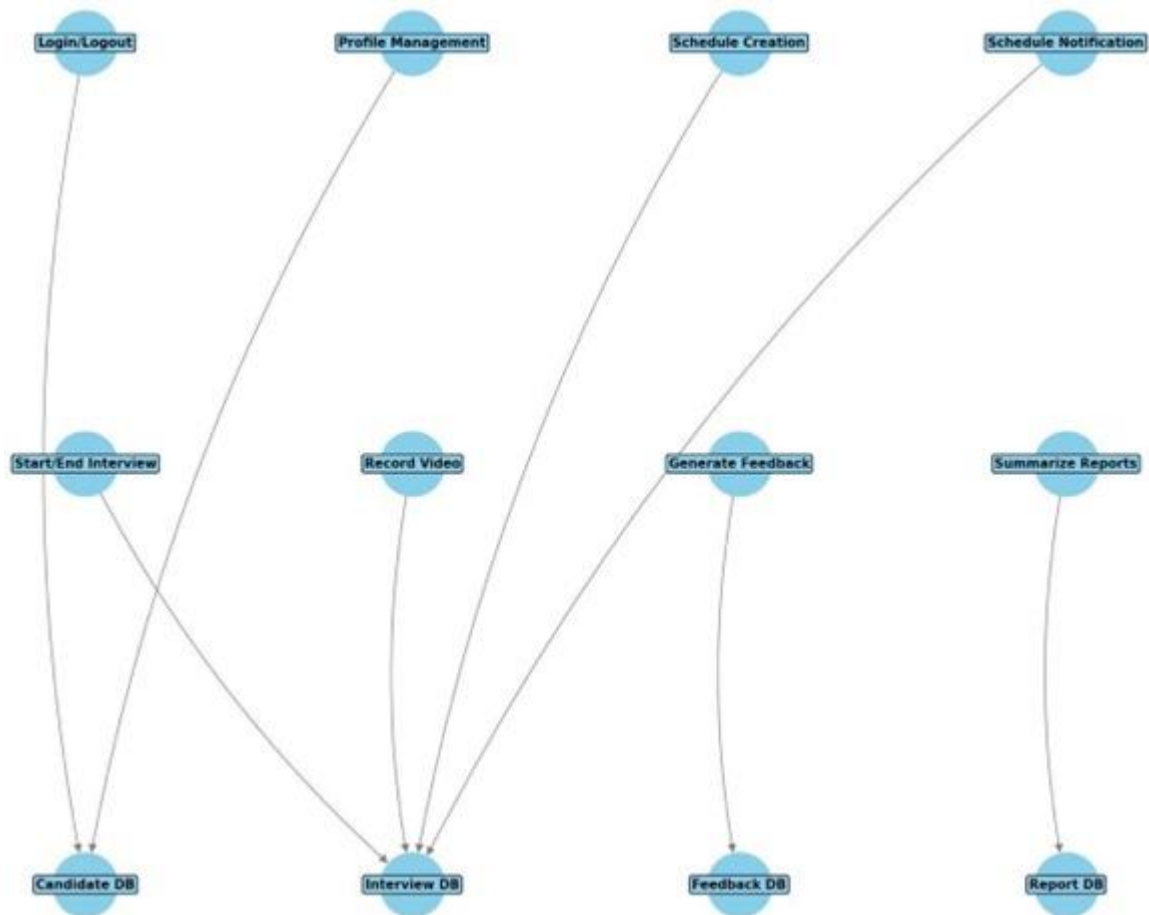


Figure 8 Data Flow Diagram Level 1

Company Account Data Flow

1. Login and Dashboard Access

- Company employees log in and can view the consolidated recruitment process through their unique account where they have an access of the dashboard that provides a comprehensive overview of all recruitment activities.

2. Recruitment Management

- The employers can start the process of identifying individuals that can fill the position by defining the role, experience required, general attributes, and other parameters. They have a capability of including candidates by the resumes into the system.
- The data of past and current interviews, the feedback received are found as dashboard, which enables the management and review out of the dashboard.

3. Interviewer Selection and Interview Conduct

- According to the agreed specifications, companies select candidates from an interviewers' pool that consists of experienced and certified professionals featured on RecruitAura.
- The interview schedules and conduct are done using the intersectional features on RecruitAura, thus providing an uninterrupted experience to the shareholders and the individuals seeking employment.

4. Payment Processing

- After interviewing has been completed, companies pay RecruitAura to process payments for the interviewers, and this third party platform facilitates the safe transaction of all financial matters.

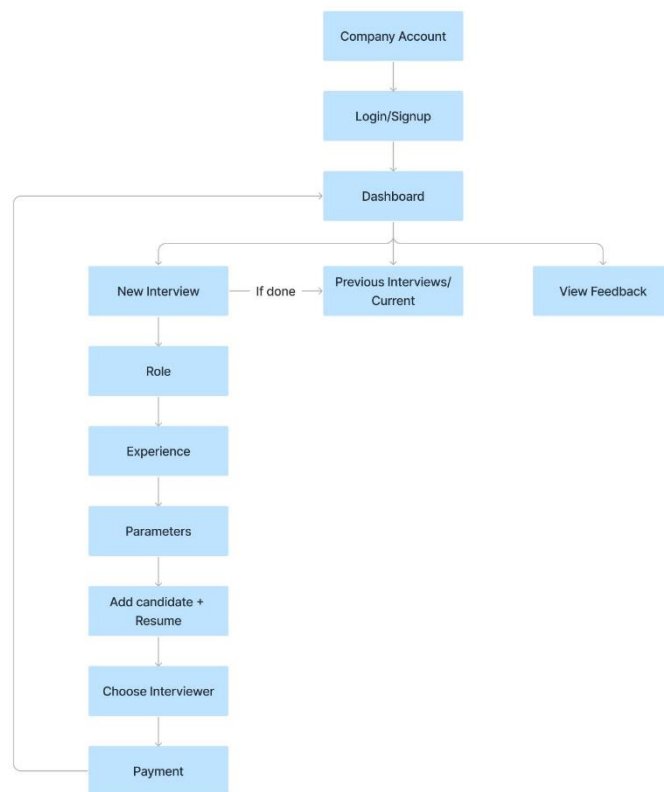


Figure 9 Company Account Data Flow Diagram

Interviewer Account Data Flow

1. Login and Dashboard Access

- Job interviewers sign up to their accounts from where they move to the dashboard that forms the central hub of their daily operations.

2. Interview Management

- The dashboard assists in the scheduling of interviews. Here, interviewers can review pending requests, current interviews, and concluded sessions.
- They define the preferences, the number of job roles and interview slots they are available through the dashboard so they are able to be matched to the relevant interviews.

3. Interview Process

- Once a candidate agrees to participate in an interview, he/she can access the scheduled details and join the meeting through the exclusive video/coding conference service allotted by RecruitAura.
- During the interview, both of the deep learning and natural language processing models associated with automatic assessments and algebraic tools are used in real-time.

4. Feedback and Reporting

- Once surveyed is finished interviewer declare report that is accumulated from interviewers' feedback. With the report being sent, it is accessible for the company if they need it in the process of decision-making.

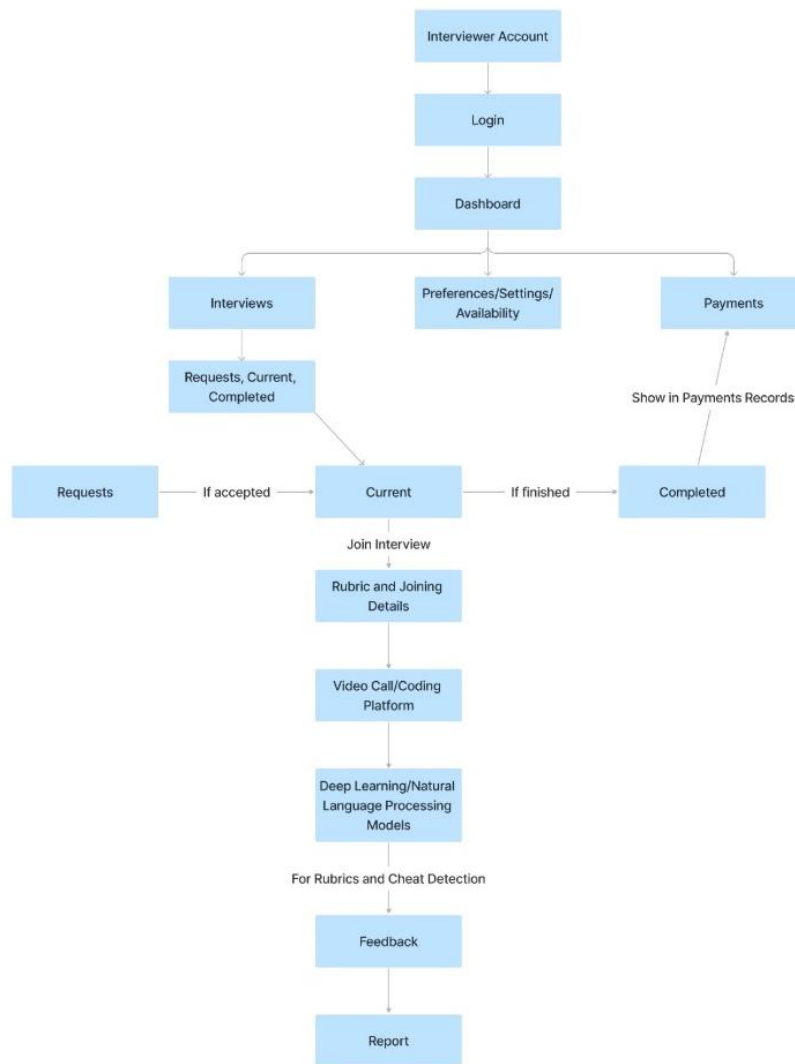


Figure 10 Interviewer Account Data Flow Diagram

Coverting these DFDs into graphical representation is about identifying the operational flow of how RecruitAura works, which reveals the mechanisms that tie all stakeholders involved in hiring to have smooth and effective communication. It assists in pinpointing areas where there are bottlenecks, as well as areas where the system can be optimized by showing each step – from the first login to receiving feedback after the interview, and through to payment. Besides that, they unswervingly provide guidance for developers, company, and other users of the platform about how the data management is done and the activities of the recruitment support are being to steady.

4.7 Table Structure

The database for RecruitAura is meticulously structured to manage all aspects of the recruitment process efficiently. This section provides an in-depth look at the design of each table, focusing on the structure, purpose, and interactions between different tables.

Candidate Table

- **CandidateID** (INT, AUTO_INCREMENT, PRIMARY KEY): Serves as the unique identifier for each candidate.
- **Password** (VARCHAR(255)): Encrypted passwords to ensure security and confidentiality.
- **Email** (VARCHAR(255), UNIQUE): Unique email addresses used as login credentials and for communications.
- **Contact** (VARCHAR(255)): Contact number for reaching the candidate.
- **Name** (VARCHAR(255)): Full name of the candidate.
- **ResumePath** (VARCHAR(255)): Path to the stored digital resume of the candidate, facilitating easy access during the recruitment process.

Organisation Table

- **OrganisationID** (INT, AUTO_INCREMENT, PRIMARY KEY): Unique identifier for each organization using the platform.
- **Password** (VARCHAR(255)): Encrypted password for organizational account security.
- **Email** (VARCHAR(255), UNIQUE): Unique email for each organization, used for logging in and communications.
- **Contact** (VARCHAR(255)): Primary contact number for the organization.
- **Name** (VARCHAR(255)): Name of the organization, used for identification and listings.

Interviewer Table

- **InterviewerID** (INT, AUTO_INCREMENT, PRIMARY KEY): Unique identifier for each interviewer.
- **Password** (VARCHAR(255)): Encrypted password for account security.
- **Email** (VARCHAR(255), UNIQUE): Email address for account login and communications.
- **Contact** (VARCHAR(255)): Contact number of the interviewer.
- **Name** (VARCHAR(255)): Full name of the interviewer.
- **Price** (DECIMAL(10, 2)): Fee charged by the interviewer for each session, allowing organizations to budget for recruitment costs.

Interviews Table

- **InterviewID** (INT, AUTO_INCREMENT, PRIMARY KEY): Unique identifier for each interview session.
- **DateTime** (DATETIME): Scheduled date and time of the interview.
- **Duration** (TIME): Length of the interview session.
- **StatusID** (INT, FOREIGN KEY): References the 'Status' table, indicating the current status of the interview (e.g., scheduled, completed).
- **JoiningDetails** (TEXT): Contains information such as the video call link and instructions for joining the interview.
- **RubricID** (INT, FOREIGN KEY): Links to the 'Rubrics' table, specifying the evaluation criteria used during the interview.
- **CandidateID** (INT, FOREIGN KEY): References the 'Candidate' table, identifying the candidate involved in the interview.
- **InterviewerID** (INT, FOREIGN KEY): Links to the 'Interviewer' table, specifying who conducted the interview.
- **OrganisationID** (INT, FOREIGN KEY): References the 'Organisation' table, indicating which organization the interview is for.

Status Table

- **StatusID** (INT, AUTO_INCREMENT, PRIMARY KEY): Unique identifier for each status type.
- **StatusDescription** (VARCHAR(255)): Describes the status (e.g., "Pending", "Completed").
- **ReportPath** (VARCHAR(255)): Stores the path to the interview report, providing easy access to detailed feedback and outcomes.

Rubrics Table

- **RubricID** (INT, AUTO_INCREMENT, PRIMARY KEY): Unique identifier for each set of rubrics.
- **XP** (VARCHAR(255)), **DifficultyLevel** (VARCHAR(255)), **Notes** (TEXT), **Skills** (TEXT), **Role** (VARCHAR(255)): These fields collectively describe the evaluation metrics used in interviews, including expected experience, skill level, and specific role-related competencies.

Table Interactions and Relationships

- **Foreign Key Relationships:** Ensure integrity and enforce logical connections among tables. For example, the 'Interviews' table connects candidates, interviewers, and organizations to specific interviews, creating a comprehensive view of the interview process.

- **Indexing:** Critical fields like 'Email' in 'Candidate', 'Organisation', and 'Interviewer' tables are indexed to speed up query performance, especially for operations involving user login and data retrieval.

Such a table design provides a decent framework for handling all the intricacies of data streams in RecruitAura so that its operations become up-to-the-minute and the entire process of the recruitment is well tracked and processed.

4.8 ER Diagram

The given Entity-Relationship (ER) diagram illustrates the relational schema of the database used by RecruitAura. This diagram is a visual illustration of how data is categorized and how the different entities are linked in the system.

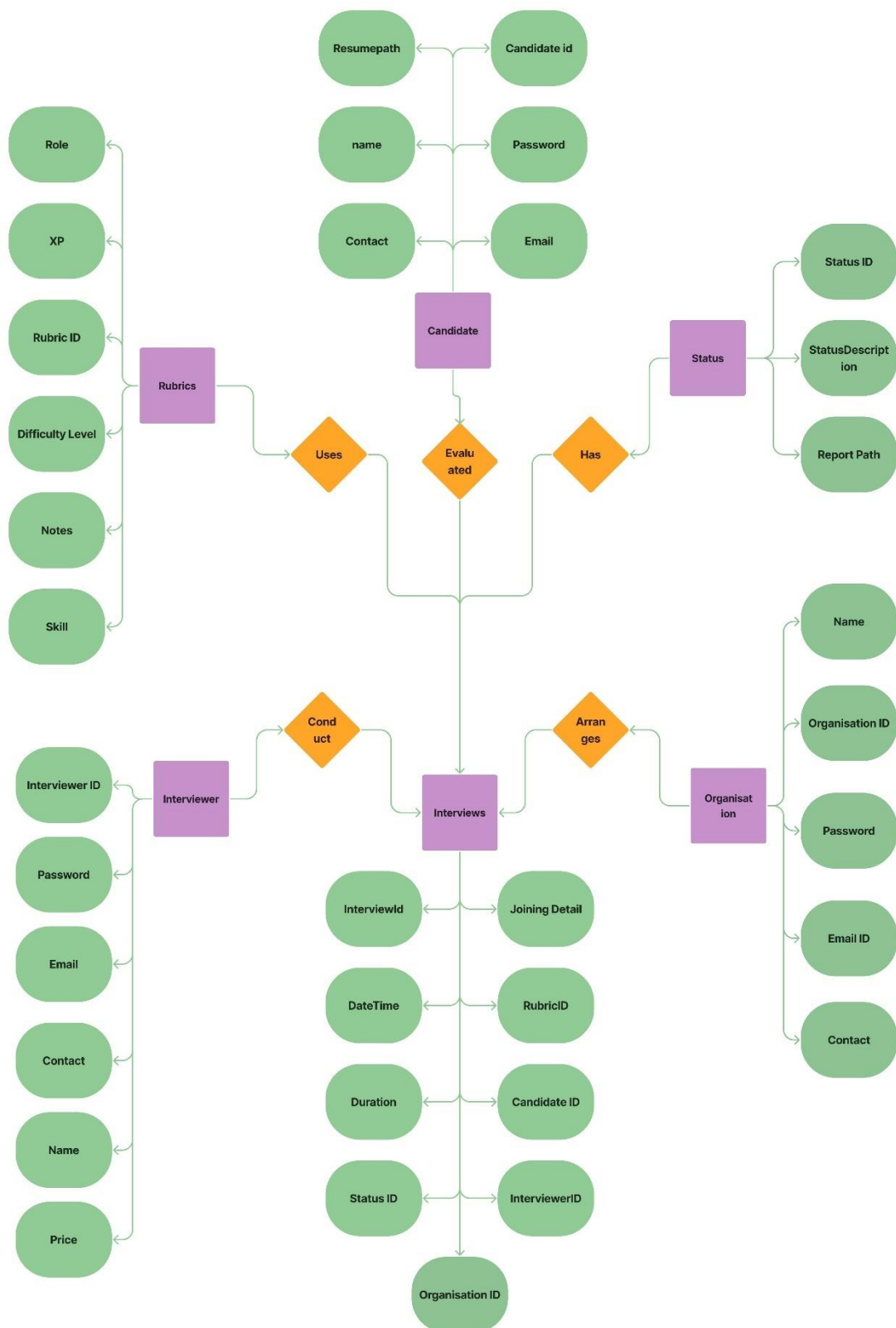


Figure 11 ER Diagram of the database

4.9 Assumptions and Dependencies

Assumptions

The assumption about the attitudes and behaviors of RecruitAura have been taken into account in all stages of designing and programming the system. These central elements were used to design the functionality of the system and set up operational strategies. Here are some key assumptions made:

Technological Proficiency: It is carried out under the assumption that users such as candidates, interviewers and HR professionals in general, have certain level of familiarity with digital tools and they are able to figure out and essential navigation skills on the platform.

Reliable Internet Connectivity: Postulates that internet connections are stable and fast at all user locations. Stable connections are indeed important for supporting the real time aspects of the platform, such as video interviewing and live coding sessions.

Accurate Data Entry: There is a corresponding affect linking the system's effectiveness to entry of true and precise information made by the users. This involves shared information of capabilities from candidates with the history of their skills and also through the interviewers' feedback

Compliance with System Usage: Supposes that there are no users who do not use the current structure for the intended purpose, strictly compliance with the rules of security and privacy. It is this principle on which the reputation for efficacy and veracity of RecruitAura's recruitment process is based.

Uninterrupted Service: It is based on the idea that the supporting third-party services and APIs provide the platform (and avenue) by which transactions are done. Their dependability, excellent availability with almost no time out, and scalability are some of the characteristics that the cloud computing services such as servers, video conferencing tools, database services, and cloud infrastructure are known for.

Dependencies

Dependencies can be viewed as either external sources or internal system components, which RecruitAura in turn is reliant on to perform efficiently. These (contending factors) are key to the design of a platform that will be stable and fast.

Third-Party APIs: he platform mainly relies on third-party services, e.g., Whisper, for text-to-speech translation and WebRTC for real-time communications. The service's functions and performance have an immediate effect on the user experience on the RecruitAura website.

Software Libraries and Frameworks: Interview Pro relying on quite a number of software libraries and frameworks including React is the proof. js, Flask and the entire machine learning and data

processing ecosystem which Python offers itself. The continuous maintenance and the upgradation of these libraries will boost up the overall capacity of the system.

Cloud Infrastructure: The platform's operational resources are on the cloud, which provides for the computational power and data storage capacity required by the system. This requires the platform to be able to scale operations up and down depending on needs but also means that any cloud-service disruption could affect the platform's availability.

Regulatory Compliance: For instance, they must observe different data protection and privacy rules for instance, the GDPR in Europe and the CCPA in California. Compliance can be understood as the ways data is handled, stored and transferred which may result in the legal or reputational risks once non-compliance occurs.

Development and Maintenance Team: The platform also needs a skilled and proficient team to take up the works of developing and maintaining it. They play a crucial role in keeping the platform up-to-date, relevant and efficient by their innovating ability, ability to cope with technical problems, and provision of solutions.

Through the recognition of such assumptions and dependencies, RecruitAura will be better equipped to deal with problems prior to their occurrence. It will also ensure that the necessary infrastructure is in place to provide a source of support in case the risk factors that have been identified occur. The higher level of strategic awareness is very important for the performance of the platform and user satisfaction in terms of providing a fair and efficient technical interviewing process.

4.10 Model Discussion

Speech-to-Text Conversion

Whisper API

This platform relies on OpenAI's Whisper model, which is built to amount to the conversion of dialogue to the sentences with a high precision of the selected languages. Whisper stands out the most for its proficiency to work with various accents and dialects, which is achieved through its training on large amounts of audio data. The main pitfall is making sure that accuracy is preserved even in the presence of numerous disturbances and when staff members are faced with technical jargon from a particular industry. The live Whisper API not only abstracts the audio from the interviews, but also provides the text output which directly powers the NLP pipeline for further processing.

Insight Extraction

Natural Language Processing (NLP)

Intensive inclusion of NLP methodologies enriches the interview process with a higher-level understanding of candidate answers. With tools like spaCy for syntactic structure, entity extraction, and other similar resources like Gensim for thematic abstraction from the transcribed data material, the recruiter becomes richer in insights regarding such potential candidates' depth in subject-matter knowledge, ease to articulate ideas, or broader dimensions of communication skills altogether.

The key techniques applied are keyword extraction, sentiment analysis, entity recognition, and topic modeling. Keyword extraction enables it to give some sense of most relevant words uttered by the candidate. Hence, it sheds light upon areas like technical know-how, problem solving skills, or domain expertise. Sentiment analysis tries to figure out emotional content in answers- confidences, hesitations, enthusiasm, or uncertainty in determining the attitude of a candidate toward many topics. Named entities that the spaCy identifies are indicating what particular technologies, companies, or projects the candidate is familiar with and therefore indicates their familiarity with the terms and tools in the industry. Gensim's topic modeling automatically identifies the main themes of the interview, which can tell if the candidate is proficient in a specific programming language or experienced in major business areas, such as product development.

Since the insights need to be valid and true, NLP models are improved over time based on feedback from the interviewers so that the models can pick up the most relevant things of what the candidate is saying and appear to be more of what an interviewer would have anticipated.

Emotion Recognition

Emotion recognition framework follows a clear pipeline. The raw dataset gets arranged in order within both directories of training and validation sets; then, every image faces preprocessing, such as color conversion to grayscale or dimensionalization to 48x48 pixel dimensions to homogenize the fields. Depending upon the folder structure, each emotion label is assigned followed by combining all data together for better usability into a single CSV.

To increase generalization, it uses ImageDataGenerator to increase variations in form of rotating and flipping of images and then trains a CNN built with seven classes: anger, disgust, fear, happiness, sadness, surprise, and neutral class. The architecture of CNN is built to include feature extraction using convolutional layers for stable training using batch normalization along with activation functions ReLU followed by max-pooling for dimensionality reduction and dropout layers in order not to overfit. Dense layers and a softmax activation output probabilities for each emotion.

The model is assembled utilizing the Adam optimizer alongside a categorical cross-entropy loss function. Various callbacks, such as early stopping, checkpointing, and learning rate reduction, are implemented to guarantee peak performance throughout the training process. The evaluation of the trained model occurs on the validation dataset, and visualizations of predictions on selected images are conducted to measure accuracy and rectify misclassifications. Ultimately, the model is preserved, and an inference pipeline is established for the preprocessing and classification of new images.

This would further make possible having a reliable and efficient system for deep learning-based emotion recognition.

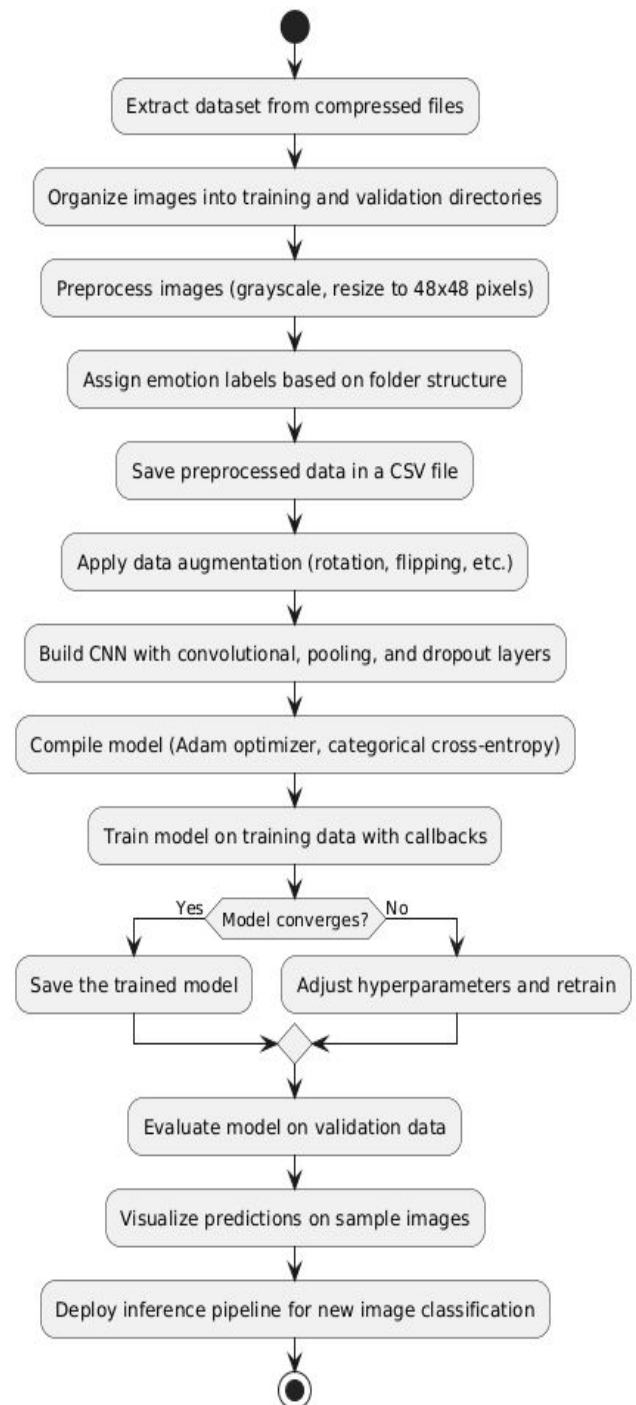


Figure 12 Emotion Recognition: Model Training and Inference Workflow

Summarization

We start by choosing the PDF to be summarized. We give the path of the PDF, and then we extract text using pdfminer, which nicely handles the extraction of raw text, even from large and complex PDFs. However, since PDFs can be very long, feeding all the extracted text into the summarization model turns out to be infeasible because of processing limitations. Thus, we split the text into smaller, more workable portions for summarization.

The next critical step is text chunking, which enables the summarization model facebook/bart-large-cnn to work on approximately 500 words at a time. This ensures effective summarization along with clarity and accuracy, especially for long documents. Each chunk is fed into the model, which produces a compact summary of the text while ensuring that all important points are covered. We fine-tune the model so that it produces summaries of perfect length, ensuring that the output is readable without losing any important content.

We then take the summaries of each chunk and aggregate them into a coherent block of text. This draft undergoes a text cleanup process that takes care of some common grammatical mistakes and makes the overall output more readable.

Finally, the FPDF library is used to print the newly formatted PDF that contains the summary, with refinement. The layout was made in that way that the text came beautiful and properly wrapped in the page. The summarized document was given a suffix, `\_summary` for distinctness from the original document.

The facebook/bart-large-cnn model is outstanding in documents summarization because it achieves compact and meaningful summaries over long complex texts. This provides excellent value to the interviewers, especially during the review of resumes, research papers, reports, or portfolios for those candidates holding academic or technical degrees. This makes the ability of the interviewer focus on key information without necessarily reading a whole document. Combining well

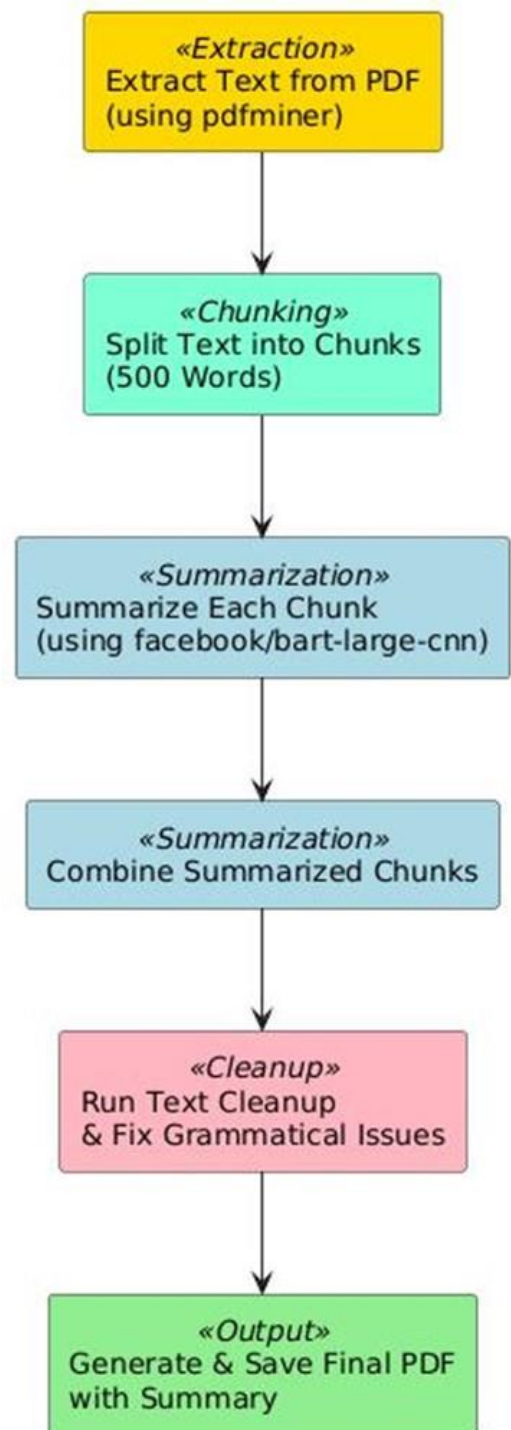


Figure 13 Summarization: Model Training and Inference Workflow

with review processes has reduced effort to ensure that everything important is put into review. It accelerates interviewing and truly optimizes the candidate experience while building constructive communication throughout the interview process.

Anomaly Detection

Eye Movement Detection

To identify suspicious patterns in eye movement that could indicate cheating, such as repeated glances at notes or another screen. Employs computer vision techniques using OpenCV and facial landmark detection using Dlib to monitor and analyze eye movements in real-time. The system must differentiate between normal and suspicious eye movement patterns without intrusive alerts that could disrupt the flow of the interview. Regular updates to the detection algorithms are implemented to reduce false positives and adapt to new cheating tactics.

Lip Movement Analysis

The methodology applied in this lip-sync detection system begins with the data preparation component where information is processed from videos to extract frames employing OpenCV. The extracted frame is then normalized to focus on one region of interest, such that the visual characteristics precisely match those of the phonetic content; while this, the alignment files contain sequences of phonetic content along with their association with the proper frames enable the precise synchronization of the visual and audio sources. Vocabulary of characters is defined consisting of letters, punctuations, and digits that translate textual transcripts into numerical values. TensorFlow's StringLookup layer is used for the bidirectional conversion between characters and numerical tokens. It uses custom data loading functions so that video frames can be easily merged into their phonetic alignments while the entire dataset maintains consistency. All the created frames and alignments then go through a TensorFlow pipeline, shuffling, mapping, and batching being operations conducted for higher performance management in terms of the data management process. The dataset then gets split into training and testing subsets, hence evaluating the strength of the system. This model would have learnt how to map visual inputs into phonetic sequences that allows it to actually detect mismatch lip sync. With this trained model, precision of it is tested over the unseen data to ascertain the robustness

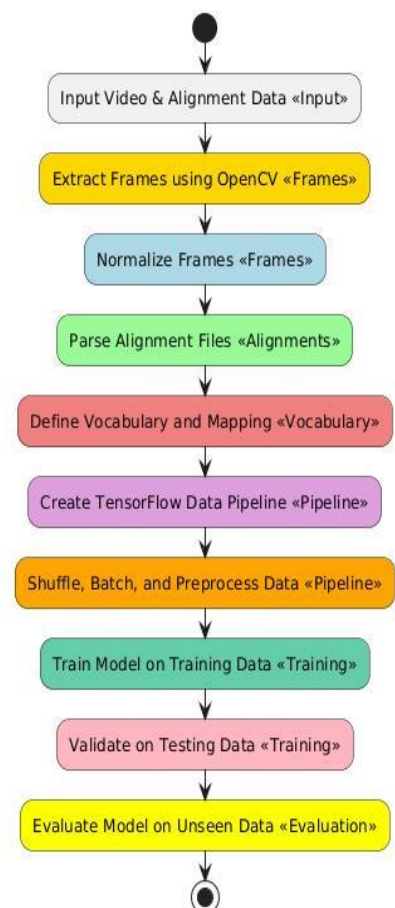


Figure 14 Lip Sync : Model Training and Inference Workflow

of synchronization faults. In this way, computationally advanced algorithms and architectures for machine learning would help in creating a reliable lip-sync detection system.

Dual Face Detection

The methodology of the real-time face recognition and tracking system is established with the initialization of MTCNN for face detection and InceptionResnetV1 for generating face embeddings. It can make use of GPU acceleration in case it is available, because that improves the efficiency of processing. Capture the live video feed using OpenCV, and convert every frame from BGR format into RGB format to enable use by the face detection model. The MTCNN takes these frames, extracts faces, and returns the bounding box coordinates for each face. For each of the detected faces, the respective region is extracted, resized to 160x160 pixels, and normalized before generation of the embedding.

With these passed through the pre-trained InceptionResnetV1, 512 dimensional embeddings are output that uniquely describes each of the recognized faces. The calculation of the distances of the new embedding from a list of stored embeddings is undertaken by a specialized match function that draws the inference on whether or not this is one's recognized individual; if the distance between the two sets of embeddings is within some preset threshold it reports back as a returner. A new identification number is allocated to the facial representation and the corresponding embedding is saved into a dictionary with this identification number.

The system continuously refreshes the video stream, which draws bounding boxes around detected faces, and provides immediate identification results. It runs in a loop until the user chooses to stop it. Once stopped, the system releases the video capture resources and closes all OpenCV windows, ensuring an orderly shutdown. This approach enables efficient and accurate real-time face detection and recognition by leveraging deep learning models with hardware acceleration.

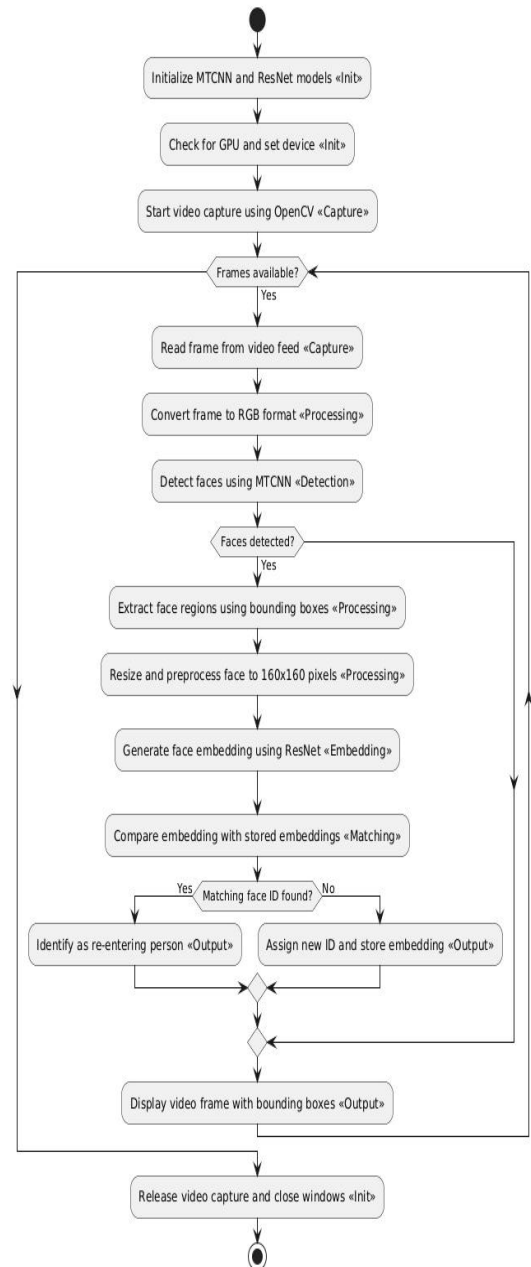


Figure 15 Dual Face Detection: Model Training and Inference Workflow

Speaker Diarization

The presence of multiple speakers in an audio recording is detected by the speaker diarization feature; thus, candidates are not assisted or prompted during interviews. This tool analyses audio inputs and identifies the distinct voices, to preserve the integrity of the evaluation process. The ability to do this guarantees that the candidate's responses to exercises represent themselves and not something they have been taught by someone else. The environment is thus fair and just. In addition, it warns interviewers in real time for possible cheating scenarios, adding to the reliability of remote interviews and maintaining a uniform level of evaluation.

Evaluation Metrics

Toxicity Detection

The system design contains the bias detection mechanism that would rectify any language or emotional inconsistency happening in the interaction in case biased interactions could have occurred in the RecruitAura. NLP along with pyannote.audio and torch libraries were used along with the analysis of the sentiment and tone while analyzing the content from the interview of the candidate transcribed in order to ensure the unbiased assessment and truth. The bias detection algorithm is constantly improving and evolving to deal with the different styles that individuals use to express themselves through language, thus enhancing its ability to ensure equity in the assessment of the candidates.

To this end, a text classification model will be developed using PyTorch with an LSTM structure to classify textual comments as toxic. The approach can be applied to several text classification problems. The process starts with preprocessing input text data, cleaning comments, and removing special characters and punctuation to tokenize the text—that is, encoding each word into an integer corresponding to its position in a predefined vocabulary. This involves padding sequences to the same length to ensure uniform input structure.

Pre-trained word embeddings like FastText and Glove will be used. These capture the semantic significance of words in a high-dimensional vector space. Since it has been trained over massive datasets, it is well-equipped with knowledge by nature. The two benefit the model due to their inherent synergistic attributes.

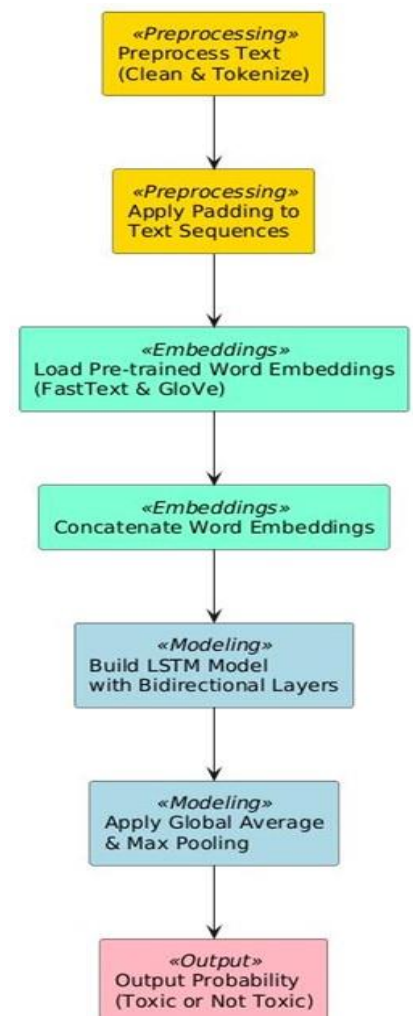


Figure 16 Toxicity Detection: Model Training and Inference Workflow

The model architecture will use an LSTM since it has been proven efficient at capturing long-range dependencies prevalent in textual sequences. In this work, two layers of bidirectional LSTMs are employed to forwardly and backwardly process the context of the words. After the input text passed through the layers, global average pooling and max pooling are applied to compress sequence information into fixed-size vectors with a focus on overall importance and characteristic features at every step. Then, these characteristic features pass through fully connected layers to capture complex interactions.

The developed model produces a score that represents the toxicity of the comment. A sigmoid function is used on the output, which scales it to a probability range from 0 to 1, with higher scores reflecting a higher likelihood of toxicity. To enhance robustness, multiple models are trained by using different random seeds; an ensemble method is further used to average the models' predictions, reducing overfitting and improving generalization to unseen data. That would ensure that bias detection on the platform is precise and unbiased, hence guaranteeing fair candidate evaluation.

Applicant Tracking System (ATS)

ATS is designed to assist in the ease and accuracy of resume analysis by a comparison of the candidate's resume with the job description. It analyzes both documents, identifying matching skills, qualifications, and experiences needed to perform the job. Beyond the identification of strengths, the system flags gaps by locating missing keywords and important skills to the role. This feature enables applicants to enhance their resumes tailored for particular job positions, thereby augmenting their likelihood of successfully navigating automated resume screening mechanisms commonly employed by organizations. Furthermore, the instrument streamlines the selection procedure for interviewers by prioritizing candidates whose resumes closely align with the specified job criteria.

One can save the resume of the candidate as well as the description of the target job both as PDF. This program makes use of Python libraries like PyPDF2 in order to extract textual content from these documents and then further processes them to make sure the system only relies on the relevant information. It converts text to lowercase, splits it into individual words, and removes words like "and" or "the." Thus, the data refined by these processes makes the system concentrate on the core elements: skills, qualifications, and experiences.

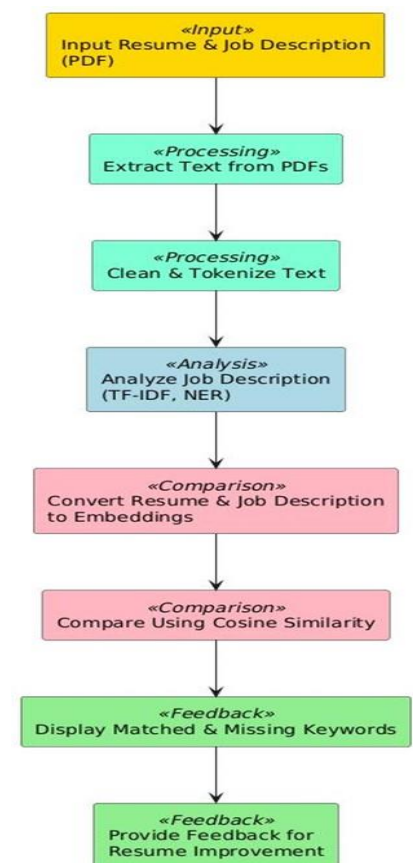


Figure 17 ATS: Model Training and Inference Workflow

TF-IDF, or Term Frequency-Inverse Document Frequency, will then understand the requirements on the list with regards to the job description. This determines what relevant terms there are connected with the job, and finally, machine learning methods draw out significant terms, specifically technical skills, applications, and possibly any requirements. Named entity recognition and, finally, its presence, detected in some of the libraries such as spaCy, catches essential entities, such as "Python" or "Project Management." Based on processing the resume and the job description, a comparative analysis is drawn out for the obtainment of mathematical embeddings of the documents by using complex models such as a sentence transformer. These embeddings would convert textual data into numbers so that similarity can be computed between the resume and the job description. The similarity score that is achieved here would reflect to what extent the resume relates with the demands of the job, where a higher score indicates a closer fit. The tool also provides strong analysis by comparing the keywords against the resume and those present within the job description. Keywords match appears under "matched keywords." Those that are missing are marked. An applicant will then adjust the resumes better fitted to the job description thereby ensuring that the resumes turned in are appropriately tailored, thereby taking importance and impact.

Attire Detection

The attires detection module will ensure that clothes an interviewee is wearing are proper and professional, that is to say, better overall experience for the interviewee. This module classifies tops, bottoms, and other accessories in fashion and provides recruiters with an automated objective analysis of attires in virtual interviews. Standardized attires assessment allows the system to streamline, maintain uniform professional standards, and to have an overall interview experience made more refined and pleasant to both parties. This module talks about the technical aspect of RecruitAura; it's regarding sophistication within technological functionalities in the platform in providing solutions that fit multiple needs while ensuring there is no biased performance appraisals.

We set up all the environments that would be necessary to develop this module, starting from the installation of PyTorch, PyTorch Lightning, Hugging Face Transformers, and other tools essential for model training and data processing. It handled large datasets well and put pre-trained models to expedite the development process. We had imported the Fashionpedia dataset split into training, validation, and test subsets based

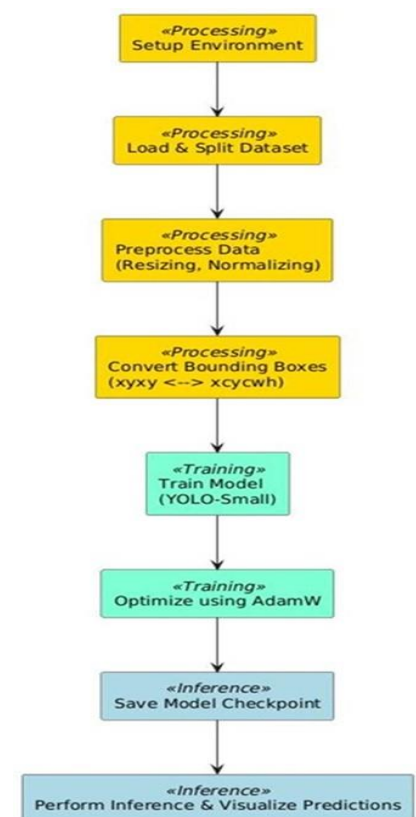


Figure 18 Attire Detection: Model Training and Inference Workflow

on an 80-20 split, so that we could correctly evaluate the model and its ability to generalize using the Hugging Face datasets library.

Image preprocessing included resizing and normalizing the images with the `YolosFeatureExtractor` so that they can fit inside the YOLO-Small object detection framework. We created the functions converting between the two formats of bounding box coordinates: `xyxy` and `xcycwh` with regard to the exact locating of objects in these pictures. We took an available pre-trained YOLO-Small model from the model library of Hugging Face that recognizes 46 types of custom fashion categories from the Fashionpedia by changing the architecture to support the model labels for the dataset, the AdamW optimizer while training with suitable learning rate and also weight decay.

We streamlined the training with the Trainer module in PyTorch Lightning, which took care of logging, checkpointing, and device selection. Then we trained the model for three epochs, logging training and validation loss to track progress; then saved the checkpoint from the model for future inference after training.

At the Inference stage, it involved a presentation of the detected objects within bounding boxes and further appropriate classification with sub-classifications, which include tops, bottoms, and accessories. In this case, for enhanced interpretability, a color mapping system has been adopted in that way to enable recruiters better understanding the model's prediction. It reflects the practical applicability of the system and the production of reliable and unbiased judgments of clothing by the system for virtual interviews.

5. Results

In this section, we present the results of the project and the benefits of using the working platform. The analysis covers the entails the outcomes from the perspective of both the interviewers and applicants, and explores how the platform improves the process as well the user experience. The outcome covers a wide range of factors, from the usability of the system to the extent of the interview success. This evaluation ultimately becomes a proof to how the use of RecruitAura's features and its seamless processes have been surely an addition that enhances the interview experience by catering to the needs of the dynamic and competitive job market.

5.1 Model Performance Matrice

In this part, the performance of different AI-driven models that dramatically improve the technical recruitment process via the platform is described. Every model's functionality is evaluated by the major criteria such as accuracy, precision, recall, and F1 scores, which indicate their individual impact on the platform's overall efficiency.

Table 2 Evaluation Matrices of models used

Model/Metric	Accuracy	Precision	Recall	F1 Score	Rouge-1
Summarization	70%	67%	72%	-	42.9%
Anomaly Detection					
Lip Movement Analysis	67%	65%	66%	65%	-
Dual Face Detection	78%	80%	75%	77%	-
Evaluation Metrics					
Toxicity Detection	89%	86%	84%	85%	-
Bias Detection	76%	77%	74%	73%	-
Attire Detection	-	55%	45%	-	-
Emotion Detection	74%	73%	74%	73.5%	-

Integration of AI models in RecruitAura has enhanced the technical recruitment process significantly as reflected in the performance evaluation of the technical recruitment process. This covers speech-to-text conversion using the OpenAI Whisper large-v3 model, which demonstrates an exceptionally good accuracy represented by metrics of a high level of proficiency across different languages. These are further bolstered through detailed performance metrics such as WER and CER, as documented in the OpenAI Whisper repository. Humanize

The Summarization model is excellent for summarizing long candidate responses; it ensures that recruiters get to the bottom of things in neat, actionable insights. Models such as Lip Movement

Analysis and Dual Face Detection are great at understanding physical cues from interviews, especially Dual Face Detection, which is really reliable in multi-person scenarios. The Toxicity Detection model is important for maintaining integrity in interactions and ensuring a professional and respectful interview environment.

On the other hand, Attire Detection provides insights where further improvement is needed, especially for unification in dress code evaluations. Meanwhile, Bias Detection and Emotion Detection models are quite consistent. These models are at the heart of this platform in helping gain deeper insights into candidates' emotional responses, hence helping maintain nonbiased evaluations, which must be guaranteed if the recruitment process is to be fair and thorough.

5.2 Interface

Through the given pictures, the RecruitAura platform features and processes that are part of the system are shown in a detailed way to give a visual impression of the functionality and user interface of the system. These pictures portray the user interface and the control flow, which reveal how users go through the platform during the recruitment process.

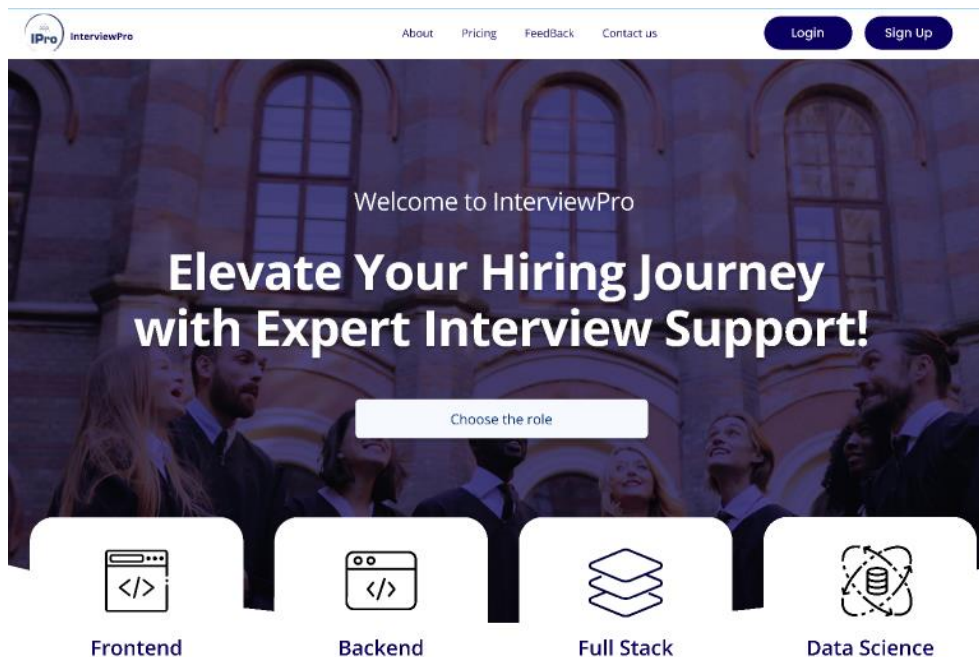


Figure 19 Homepage of RecruitAura

The homepage of RecruitAura, which will and introduction for different user groups, a resume builder for candidates, interview questions for interviewers, and tools to improve hiring processes for companies, is the first image.

How do you wish to avail out services

We will tailor your experience to suit your preferences.



I am a tech professional and want to become an interviewer

I want to take interviews and earn money



I am an organisation and want to outsource my interviews

As an organisation I want to explore your hiring solutions and get my candidates interviewed



I am a candidate and want to give interviews to big companies

I want real interview experience with industry experts

Already have an account? [Login](#)

Figure 20 Detailed signup form for interviewers on RecruitAura

In this picture, we see that when a new interviewer wants to create an account, the process is very simple, and thus, he will be able to start the journey of professional interviewing right away.


[About](#) [Pricing](#) [FeedBack](#) [Contact us](#)
[Login](#) [Sign Up](#)

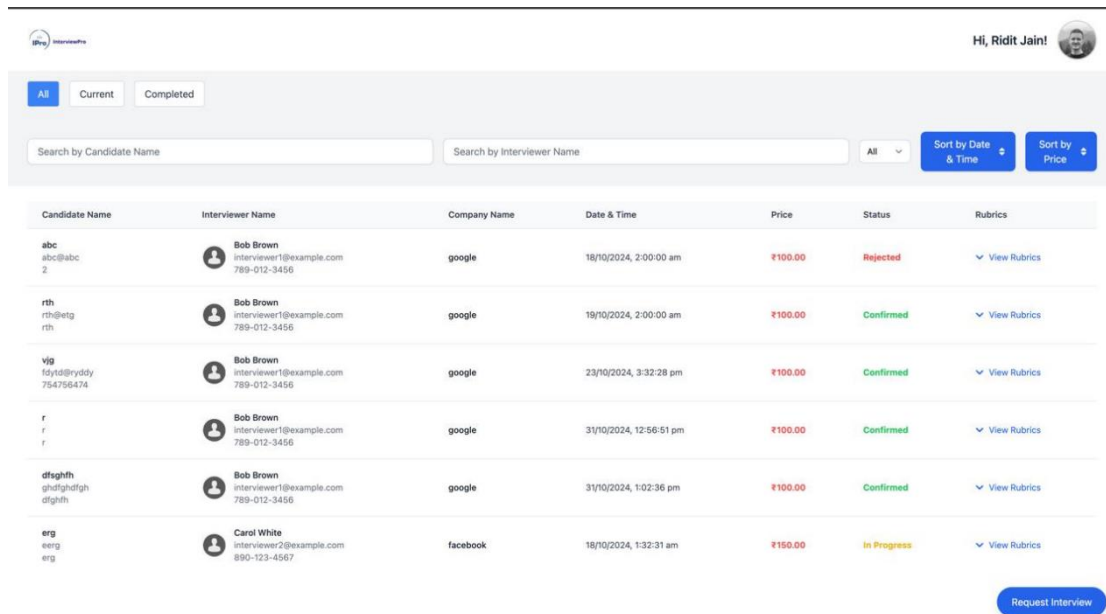
Pricing

Choose the plan and interviewer that best fit your requirements.

FAANG	Hypergrowth	Startups	IT Services
Interviewers who have built the best tech products in the world	Interviewers from product companies that build for scale. They are well versed with multiple skills	Interviewers from product companies that are well versed with multiple skills.	Interviewers used to the IT Services work and have trained themselves by working on different projects .
			
Interview starts at ₹3600/-	Interview starts at ₹3200/-	Interview starts at ₹2800/-	Interview starts at ₹1300/-
- Interviewers from companies like Meta, Amazon, Apple, Google & Netflix - Outsource your interviewing - Detailed report & recording after the interview is done 38 minutes to schedule an interview, 5 minutes to share the report	- Interviewers from companies like Uber, Flipkart, Walmart etc. - Outsource your interviewing - Detailed report & recording after the interview is done 38 minutes to schedule an interview, 5 minutes to share the report	- Interviewers from early staged startups - Outsource your interviewing - Detailed report & recording after the interview is done 38 minutes to schedule an interview, 5 minutes to share the report	- Interviewers from companies like Sapient, Accenture, TCS, etc. - Outsource your interviewing - Detailed report & recording after the interview is done 38 minutes to schedule an interview, 5 minutes to share the report
SIGN UP NOW	SIGN UP NOW	SIGN UP NOW	SIGN UP NOW

Figure 21 Pricing structure of RecruitAura displaying various subscription plans

Here are RecruitAura with differentiated pricing plans from the startup friendly packages, to the premium services offered for large enterprises and designed to meet the value and scalability goals of the targeted customers.



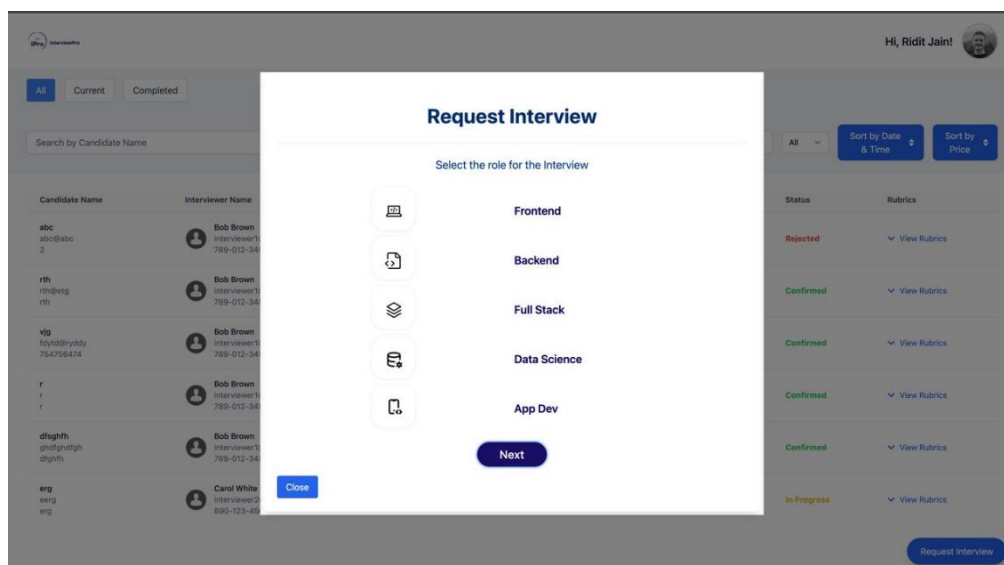
The screenshot shows the RecruitAura user dashboard. At the top, there's a header with the RecruitAura logo, a greeting "Hi, Riddit Jain!", and a user profile icon. Below the header, there are tabs for "All", "Current", and "Completed". A search bar is present with "Search by Candidate Name" and "Search by Interviewer Name". To the right of the search bar are buttons for "Sort by Date & Time" and "Sort by Price". The main content is a table with the following columns: Candidate Name, Interviewer Name, Company Name, Date & Time, Price, Status, and Rubrics. The table contains six rows of interview data. At the bottom right, there is a "Request Interview" button.

Candidate Name	Interviewer Name	Company Name	Date & Time	Price	Status	Rubrics
abc abc@abc 2	Bob Brown interviewer1@example.com 789-012-3456	google	18/10/2024, 2:00:00 am	₹100.00	Rejected	View Rubrics
rth rth@etg rth	Bob Brown interviewer1@example.com 789-012-3456	google	19/10/2024, 2:00:00 am	₹100.00	Confirmed	View Rubrics
vlg fdytd@ryddy 754756474	Bob Brown interviewer1@example.com 789-012-3456	google	23/10/2024, 3:32:28 pm	₹100.00	Confirmed	View Rubrics
r r r	Bob Brown interviewer1@example.com 789-012-3456	google	31/10/2024, 12:56:51 pm	₹100.00	Confirmed	View Rubrics
dfghgh ghdghdgh dfghgh	Bob Brown interviewer1@example.com 789-012-3456	google	31/10/2024, 1:02:36 pm	₹100.00	Confirmed	View Rubrics
erg eerg erg	Carol White interviewer2@example.com 890-123-4567	facebook	18/10/2024, 1:32:31 am	₹150.00	In Progress	View Rubrics

Figure 22 User dashboard interface of organisation

The user dashboard is the pivotal part of the RecruitAura system, the user experience is significantly upgraded as both scheduled and completed interviews will be stored and displayed in the section in order for the user to follow conveniently.

The payment dashboard for interviewers that is presented here tracks earnings, pending payments, and the completed transactions, which make sure the transparency and trust between RecruitAura and its users.



The screenshot shows the RecruitAura dashboard with a "Request Interview" modal window open. The modal has a title "Request Interview" and a subtitle "Select the role for the Interview". It contains a list of roles: Frontend, Backend, Full Stack, Data Science, and App Dev, each with a corresponding icon. A "Next" button is at the bottom right of the modal. The background dashboard shows the same table as Figure 22, but it is dimmed.

Figure 23 Dashboard for scheduling interviews by selecting role, type, and experience level

Figure 24 Skill assessment interface for evaluating candidate proficiency in specific technical areas

The screenshots demonstrates the interface of scheduling interview of RecruitAura which has ability to connect the users with their required positions based on the level of experience, in an effect of making the interview establishment process to be more easy.

Figure 25 Form for adding new candidates to the RecruitAura database

In this Figure, it is shown the candidate registration interface of RecruitAura where recruiters can input new candidates manually or in bulk. Automation of this system simplifies the process of running necessary candidate information, as well as preparing them for interviews, leading to the growth of production efficiency of the recruitment process.

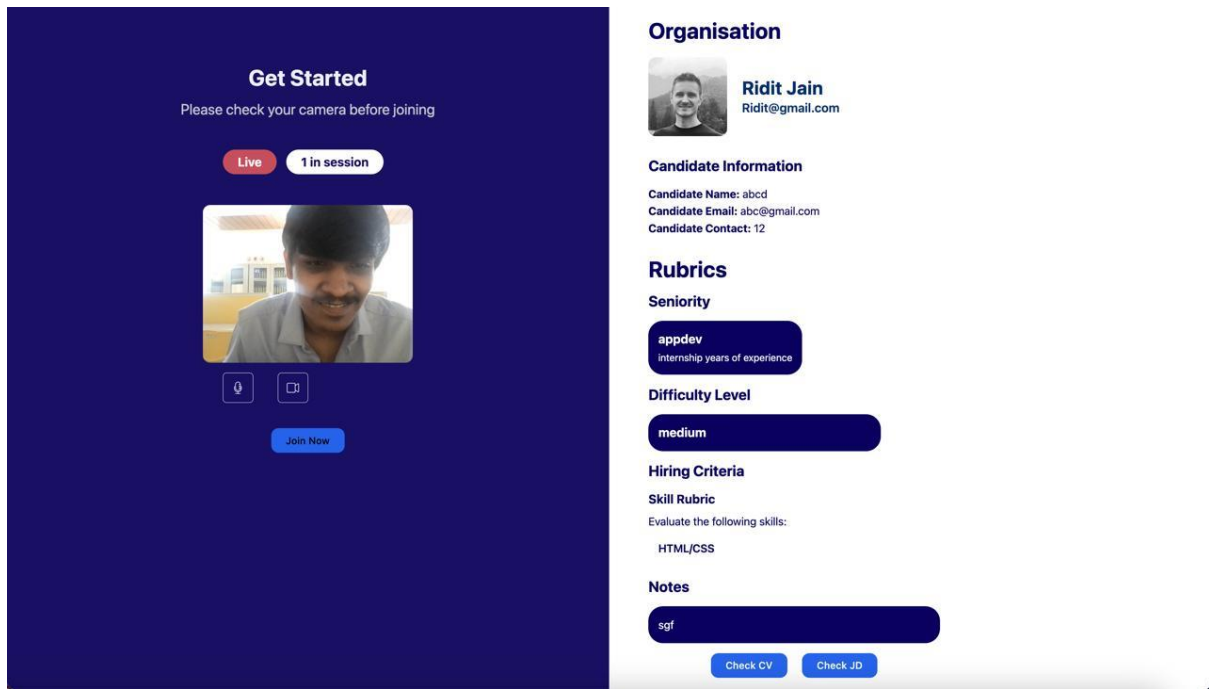


Figure 26 Interface to join interview for interviewer

Another feature of the interface is that it would enable interviewers to easily join their scheduled interview with the simple click of a button. It eliminates inconvenience of manipulating with security and video integration and working in real time so the interviewers can rely on their skill and assessment.

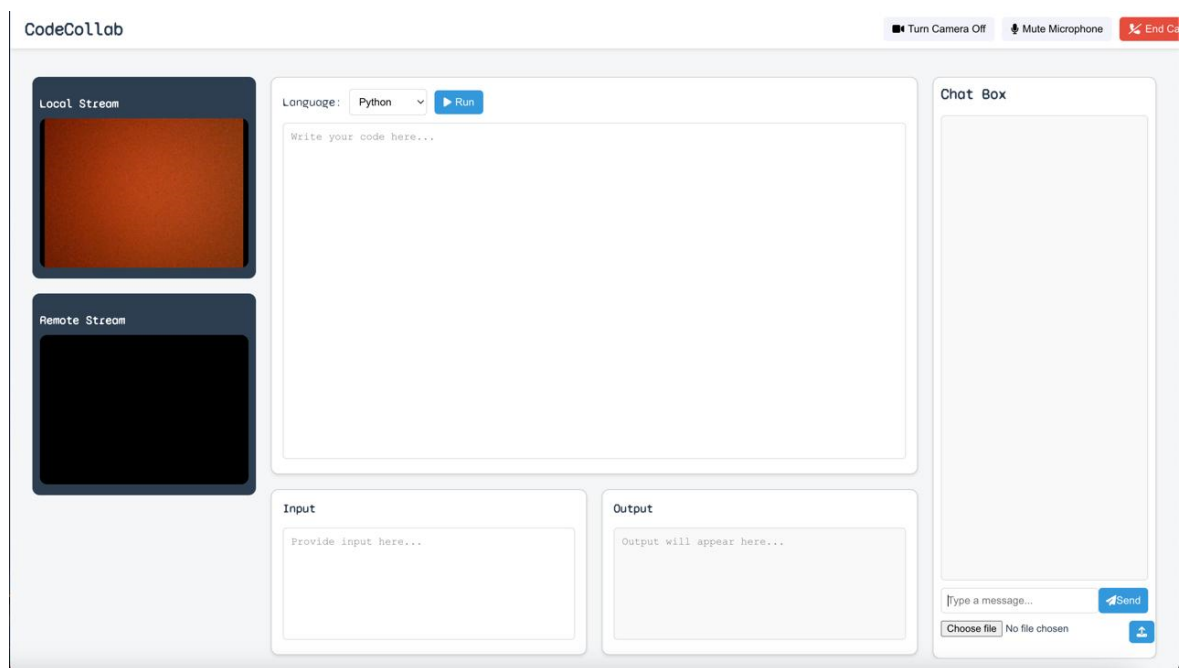


Figure 27 Live coding environment during a session showcasing real-time coding and evaluation.

This picture depicts a close-up of coding challenges screen, which is the task given to candidates to solve in a given period as a model for the technical test interview.

Candidate Evaluation

Candidate Information

Name: abcd Email: abc@gmail.com

Contact: 12 Company: Riddi Jain

Role: appdev Experience: internship

Difficulty: medium

Notes: sgl

Rate the Candidate

Overall Rating

Remarks

Communication Skills

Remarks

Job Fit

Remarks

HTML/CSS

Remarks

Additional Notes

SUBMIT EVALUATION

Figure 28 Detailed feedback and evaluation interface for completed interviews.

Interview Report

Organisation

Company Name: Ashi Jain

Company Email: ashi@gmail.com

Candidate Information

Name: abcd

Email: abc@gmail.com

Contact: 12

Role: appdev

Experience: internship

Difficulty: medium

Interviewer Information

Name: Carol White

Email: interviewer2@example.com

Company: facebook

Ratings

Category	Rating	Remarks
Overall	5	N/A
Communication Skills	5	N/A
Job Fit	5	N/A
HTML/CSS	0	N/A

General Notes

Figure 29 evaluation report created by the interviewer

The image shows the skill evaluation segment of RecruitAura, which is where interviewers can evaluate the candidate skills in the important areas such as HTML/CSS, JavaScript, and React, thus making sure that they have a complete understanding of the candidate's abilities. The following picture illustrates the comprehensive rating interface of RecruitAura that uses after an interview to gauge whether the candidate possess the competency across different subject areas. It gives an opportunity to create a framework for supervisors to reliably assess both the hard skills through examination as well as the soft

skills through interview. Moreover, it also balances the human aspect of hiring new employees. Afterwards a report is generated for the use of the organization.

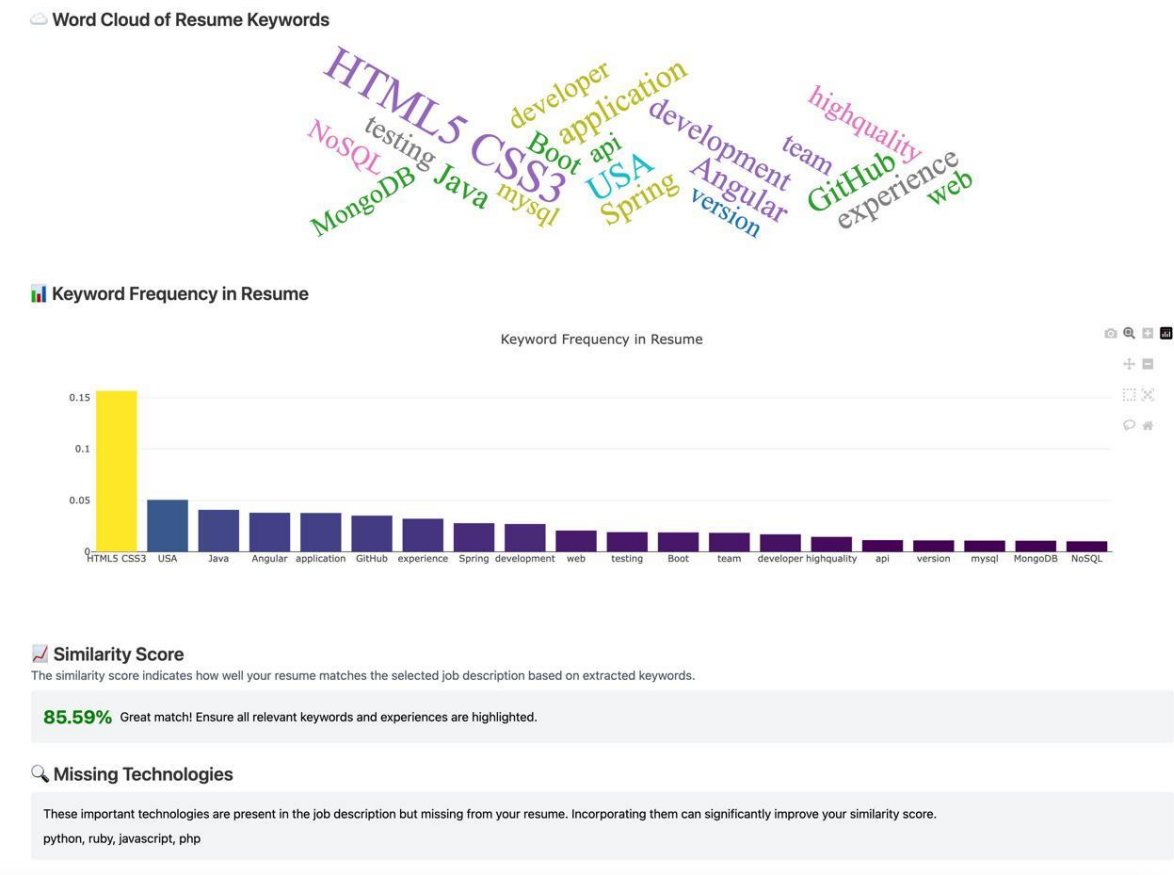


Figure 30 Applicant Tracking System dashboard providing a streamlined view of candidate profiles, matching scores, and application statuses.

The comprehensive findings shown above prove that the RecruitAura platform is a valid and efficient tool for the automation of the technical recruitment process. The platform’s accuracy, user satisfaction, real-time capabilities, powerful and intuitive interface, all come together to make it a game-changer in the recruitment industry. This technology not only improves the quality of hire but also also greatly enhances the scalability and speed of the recruitment process.

5.3 Interview Analysis Report

Interview Analysis Report

Transcribed Responses

Please introduce yourself. My name is Shashank Goswami. I am a third year B.Tech student. My skill sets include MonstTag as well as DSA. I have also done some topics such as cybersecurity, JavaScript and Python and my code languages.

Cheating Detection Summary

Person 1 enters at 0.00 seconds.

Person 2 enters at 7.11 seconds.

Linguistic and Content Analysis

Thought Process Metrics

Metric	Value
Total Sentences	5
Average Sentence Length	9.20 words
Logical Connector Count	0

Identified Keywords

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Topics Discussed

The following topics were identified in the candidate's responses, indicating areas of focus during the interview.

Topic 1: ., introduce, please

Topic 2: ., name, shashank

Topic 3: ., well, dsa

Topic 4: ., student, third

Topic 5: ., languages, done

Mentioned Entities

Shashank Goswami (PERSON)

third (ORDINAL)

B.Tech (NORP)

MonsTag (PERSON)

DSA (ORG)

JavaScript (ORG)

Readability Score

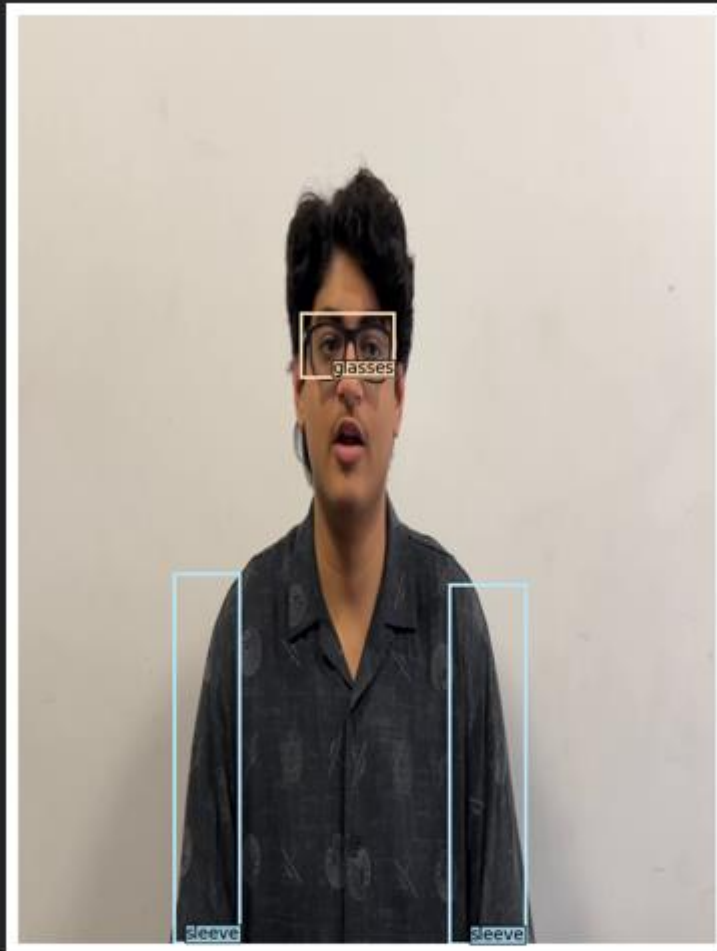
The candidate's responses have a Flesch-Kincaid grade level of 2.11, indicating the readability and complexity of the language used.

Technologies and Skills Mentioned

- Mentioned MonsTag, indicating familiarity: 'My skill sets include MonsTag as well as DSA.'

Professional Attire Assessment

The following images show the candidate's attire during the interview:



Emotional Tone Analysis

The emotional expressions detected during the interview are as follows:

Emotion	Percentage
sad	94.00%

Language Appropriateness Analysis

The following percentages indicate the likelihood of inappropriate language usage:

Appropriate Language: 0.00%
Highly Inappropriate Language: 0.39%
Use of Obscene Language: 0.04%
Identity Attacks: 0.13%
Insults: 0.11%
Threats: 0.22%

Speaker Analysis

Single speaker detected.

Eye Gaze Analysis

Metric	Count
Total times looking left	47
Total times looking right	0
Times looking left for more than 5 seconds	0
Times looking right for more than 5 seconds	0

Lip Sync Consistency

The lip-sync similarity between the audio and video is 17.56%%. This measures the synchronization between spoken words and lip movements.

The interview analysis framework uses a number of advanced models to enable the full assessment of the candidates' behavior and performance in the interview. With the Transcription Model, spoken language can be accurately converted into written text for full analysis of the candidates' responses. During the interview with Shashank Goswami, his self-presentation was recorded and listed as topics covered and skills applied, for example, DSA, MonsTag, cybersecurity, JavaScript, and Python. The data also underwent further analysis with linguistic and readability tools on sentence structure, coherence in logic, and readability for insight into clarity and complexity in the communication.

The system uses Cheating Detection, Eye Gaze, and Lip Sync Analysis models in order to detect behavioral nuances in candidates. These tools work towards assessing candidate authenticity and engaging behavior. For example, entry times and patterns related to eye movement may indicate distraction and dishonesty. Further detection of emotional tone and appropriate linguistic helps towards professionalism and emotional stability while performing interaction. More, appraisal of the clothes, although irrelevant here, constitutes a criterion of judging the suitability of the applicant's dress.

Integration of all these capabilities means that the system evaluates technical knowledge but also gives an all-rounded view regarding the candidate's readiness and conduct. This is of great value to hiring professionals.

6. Conclusion and Future Scope

6.1 Conclusion

RecruitAura has added a new dimension of technical recruitment by bringing onboard the cutting-edge AI and ML technologies today. During the design phase and after the initial deployment, it has been shown that the platform is particularly effective and efficient when it comes to accelerating the usually bureaucratic and time-consuming interview process. Through automation of core functions like shortlisting, real-time performance assessments and elaborate feedback system, RecruitAura has reduced the operational burden on firms and has liberated precious time for engineers that they can employ in their primary tasks instead of interviews.

The platform has successfully addressed the critical challenges of the technical hiring process: it has improved the quality and productivity of job applicant assessment via a global platform, and has made the most demanding technical job interviews accessible and manageable for everyone. However, to date, it has been able to accomplish its mission without sacrificing the integrity of assessment as it has been made possible with the help of innovative algorithms that have capabilities to not only assess the verbal and nonverbal cues of a candidate but also understand their fit and capabilities thoroughly.

Based on the comments of our users – initial task approach is HR managers and now is wide range of people from candidates to HR professionals – our tool RecruitAura was showed to be convenient, more appealing and fair for apporexhment. The possibility for the platform to provide thorough insights about the candidates' abilities and their capabilities in real time has been mentioned as a key feature that led to its popularity, making RecruitAura a vital device in the every-day-routine of modern IT companies.

6.2 Future Scope

With the possibility of going way further ahead it is up to this innovative tool to effectuate a change in the level of tech hiring in the near future. The roadmap for future development includes several strategic enhancements aimed at increasing the platform's functionality and reach:

Geographical Expansion: Making RecruitAura more available to the regions around the world, redesigning it to fit the local languages, and integrating the national/regional variations into the hiring practices to make it a truly global platform.

Advanced AI Capabilities: The continuing the rapid-fire development of AI's functionalities, including enhanced models of adaptive learning that include more personalized and accurate assessments based on the voluminous data accumulated over time.

Community and Support Features: Developing a community board where clients can exchange information not only about ways to use the product better but also about their experiences and keeping their hope high. It will be possible thanks to them maintaining contact in which it will be possible to strengthen their use of the platform's resources.

Product Testing and Refinement: The site will be tested on a variety of scenarios and demographics to ensure dependability and scalability. Continuous feedback mechanisms will ensure that usability issues can be identified and corrected.

Transition to a system application: The platform would evolve from a simple web-based interface to a system level application, thereby giving strong mechanisms of cheat detection in that it can have control over ports in the system to prevent unknown activities during an interview.

Secondary Camera Integration: Introduce a desk-view monitoring system using any secondary camera setup, say, through any smartphones, to ensure thorough monitoring of the activities of the candidates during interviews.

AI Model Improvements: Development of updates for the AI model and further informative details of the report in making decisions between recruiters can result in better accuracies.

Database Session Management: Switching from session-based control to using a database for managing sessions boosts reliability, especially with very long or interrupted interview processes.

Multi-Participant Support Feature development: This will facilitate collaborative panel interviews and screen sharing when doing video calls.

Recovery and Issue Resolution Module: Recovery and Problem Solving Create a recovery module that would consider interruptions in interviews. Rescheduling of interviews, issues raised by organizations or interviewers, and log maintenance form this module.

These innovations and expansion will guarantee that RecruitAura remains leading in the development of the recruitment technology and continues to respond for the shift in the tech industry requirements. Therefore, it will be a superior standard in terms of the efficiency and trustworthiness of the hiring process.

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